πPLN:

Paraconsistent PLN Without a Global Probability Space

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May 27, 2026

Abstract

The νPLN formalization of [1] gives a useful rigorous probabilistic idealization for PLN-style inference by introducing a global probability space $(\Omega, \mathcal{F}, \nu)$, a random model/predicate generator M, and observation variables D_x. The present paper does not reject this mathematics. It argues that the global space should be understood as one possible theoretical idealization, and that the more useful persistent semantics for an AGI-scale PLN system is not a single posterior belief state but a paraconsistent, context-indexed evidence metagraph. When a particular inference demands Bayesian coherence, the system temporarily constructs a local νPLN-style chart $(\Omega_C, \mathcal{F}_C, \nu_C)$, reasons inside it, and projects the result back into persistent positive and negative evidence. Gluing local charts into a larger probability model is an achievement to be earned by compatibility conditions, not an axiom of the whole architecture.

We call this view πPLN, for paraconsistent PLN. Its main truth-value object is not a scalar probability and not primarily a second-order probability distribution, but an empirical evidence object (n^+, n^-) living in an evidence-count quantale. PLN simple truth values, beta posterior truth values, and the local νPLN truth distributions of [1] are projections of this object under context-specific assumptions. Priors are therefore local projection parameters, not contents of the persistent metagraph and not global semantic foundations.

The paper also gives an explicit account of revision, because revision is the central pressure point for any non-global approach. πPLN revision is not global Bayesian conditioning. It is context-indexed evidence transformation: empirical evidence is added, discounted, corrected, quarantined, translated, or reprojected with provenance and context boundaries intact. In residuated quantales this becomes a general conservative revision rule: project candidate states and evidence into a shared context lattice, measure residual fit and residual preservation there, and choose the weakest adequate revised state. Bayesian conditioning and beta-binomial updating are recovered as special local chart projections, but they are not the definition of revision. Quantale weakness then supplies the generalized Occam principle that guides context construction, chart selection, rule scheduling, revision, and factor-graph message passing. The resulting architecture preserves the local mathematics of νPLN while avoiding the need to make a single global probability space the foundation for heterogeneous, conflicting, resource-limited, task-oriented AGI reasoning.

Executive summary for busy skimmers

The core claim is simple:

$$\nu\text{PLN} = \text{local Bayesian coherence for PLN},$$

*Warm thanks to Nil Geisweiller, whose νPLN draft [1] prompted this work and whose careful formalization of the global-probability-space view supplied the conceptual foil against which the present paper is written.

$$\pi\text{PLN} = \text{global paraconsistent contextual intelligence using local Bayesian charts.}$$

The νPLN formalism of [1] introduces a global probability space $(\Omega, \mathcal{F}, \nu)$ and treats observations as indexed Boolean random variables generated by an unknown probabilistic predicate or program. This is a clean way to derive local formulas such as the beta posterior for predicate direct introduction. The problem is not the local mathematics, nor even the use of a global space as an idealized mathematical device. The problem is making that device the only foundational semantics for PLN, rather than treating it as one chart in a larger context-indexed evidential architecture. In πPLN, gluing local charts into a larger coherent probability model is an achievement, not an axiom.

Historical PLN already contains the better philosophical instinct: truth values are context-relative, PLN does not axiomatically assume one sample space for all probabilities, and inference control is a central problem rather than a minor implementation detail [3]. The paraconsistent foundations of PLN then clarify how positive and negative evidence can be stored separately as p-bits, with PLN simple truth values recovered by renormalization [2]. Weakness theory gives the generalized inductive principle: prefer the weakest adequate hypothesis or chart, where weakness is represented in an appropriate quantale rather than reduced to a single Solomonoff prior [4]. Hyperseed's treatment of tasks, agency, observer-relativity, and paraconsistent evidence supplies the broader AGI setting in which inference is not passive prediction but context-sensitive competence in task families [5].

πPLN has two layers.

1. The persistent base layer is a paraconsistent metagraph:

$$K_t(\phi, C) = (E^+_{\phi,C}, E^-_{\phi,C}, \text{Prov}_{\phi,C}, \text{Attn}_{\phi,C}, \ldots).$$

   It stores support and opposition separately.

2. The local inference layer constructs temporary charts:

$$\Gamma_C(K_t, T) = (\Omega_C, \mathcal{F}_C, \nu_C, M_C, D^C, \iota_C),$$

   runs νPLN-style Bayesian or PLN rule calculations inside C, and then projects the result back into p-bit/count evidence.

The main truth quantale is the evidence-count quantale

$$Q^{\text{cnt}}_{\pm} = ([0, \infty]^2, \leq, \vee, +, (0, 0)),$$

whose elements are pairs (n^+, n^-) of positive and negative empirical evidence counts. It projects to a context-specific PLN SimpleTruthValue by injecting the prior only at projection time:

$$s = \frac{n^+ + kp_0}{n^+ + n^- + k}, \qquad c = \frac{n^+ + n^-}{n^+ + n^- + k}.$$

Equivalently, setting $\alpha_0 = kp_0$ and $\beta_0 = k(1 - p_0)$ gives the beta posterior projection

$$(n^+, n^-) \mapsto \text{Beta}(\alpha_0 + n^+, \beta_0 + n^-).$$

Thus the predicate direct introduction calculation of [1] is recovered exactly inside one local context, while the persistent metagraph stores empirical counts rather than prior-smeared beliefs.

Revision is the point where this architecture most sharply differs from a global-probability semantics. πPLN does not say that priors are never useful; it says that priors are not required for persistent empirical revision. The base metagraph can revise raw context-indexed evidence

$$(\phi, C, \mathrm{Prov}, E^+, E^-)$$

without first projecting it to a beta posterior or embedding it in a universal sample space. When a local Bayesian chart is later constructed, a local prior (p_0, k) or (α_0, β_0) may be injected at projection time.

The general revision schema uses residuation. If H is the current local hypothesis, E is new evidence, and K ranges over candidate revised states, define a shared lattice of evaluable propositions $\mathcal{P}_C$ and projection maps

$$\mathrm{eval}_{\mathcal{K}} : \mathcal{K}_C(H, E) \to \mathcal{P}_C, \qquad \mathrm{eval}_{\mathcal{E}} : \mathcal{E}_C \to \mathcal{P}_C.$$

Then residual fit and preservation are

$$\mathrm{fit}_C(K; E) = w_C(\mathrm{eval}_{\mathcal{E}}(E)) \Rightarrow_C w_C(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E)),$$

$$\mathrm{pres}_C(K; H) = w_C(\mathrm{eval}_{\mathcal{K}}(H)) \Rightarrow_C w_C(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{K}}(H)).$$

Hard revision maximizes preservation and weakness subject to a fit threshold; soft revision lexicographically maximizes fit, preservation, and weakness. Thus revision becomes "fit the new evidence, preserve what still works, and make no unnecessary new commitments." Bayesian conditioning is recovered only in the probability-valued special case.

The practical consequence is significant. A νPLN chainer tends to ask: "What posterior update follows from the global model?" A πPLN chainer asks: "Which local chart, under which assumptions, for which task, gives useful evidence, and how should this evidence be stored without erasing incompatible evidence from other charts?" In factor-graph terms, πPLN uses quantale-valued factor graphs with local Bayesian islands, not one monolithic probabilistic graph.

Contents

1 Introduction

PLN, Probabilistic Logic Networks, was designed as a practical, heterogeneous logic of uncertain inference suitable for integrative AGI. It has always combined several ingredients: term logic, probabilistic truth values, weight-of-evidence, heuristic independence assumptions, intensional and higher-order inference, and explicit inference control [3]. It was never merely a textbook probability calculus over a fixed sample space.

The νPLN formalism of [1] aims to give PLN a rigorous global probability-theoretic foundation. The resulting formalism is useful. It clarifies how direct introduction and related PLN rules may be derived from a Bayesian second-order distribution over possible predicate-generating programs. The central foundational move is to use one global probability distribution ν over possible worlds and histories as the mathematical basis for the definition. This is a defensible idealization, much as Solomonoff induction uses an uncomputable universal prior as an idealization. The issue is scope: it is not the only useful idealization, and it is not the right persistent semantics for a heterogeneous AGI knowledge metagraph.

The present paper relocates the global-distribution assumption. It says:

> A global probability distribution is often useful as a temporary local fiction or as a special theoretical idealization. It should not be required as the foundational semantics of the whole cognitive system. When local charts glue, use the gluing; when they do not, record the obstruction.

The alternative proposed here is πPLN, paraconsistent PLN. The name is chosen to emphasize that the enduring semantics of the system is paraconsistent, positive-negative evidence in a metagraph, while ordinary probability theory is applied locally and provisionally.

A related point concerns revision. A global Bayesian treatment can revise by conditioning one prior distribution on one accumulating body of evidence. That is elegant when the global distribution is available and the evidence is already expressed in its sample space. In πPLN, however, revision usually occurs before any such chart exists. The system receives evidence packets from sensors, documents, chainers, subagents, memories, and old proof trails. These packets may overlap, contradict, age, or use different concept boundaries. Revision must therefore be an operation on context-indexed empirical evidence with provenance, not only an operation on posterior probabilities. Local Bayesian updating remains one important projection of this process.

1.1 The guiding metaphor

A local probability space is like a coordinate chart on a manifold. It is precise and indispensable in a neighborhood. But the existence of useful local coordinates does not imply that the whole manifold is globally Euclidean.

Similarly, the νPLN probability space

$$(\Omega, \mathcal{F}, \nu) \quad \text{of [1]}$$

is valid inside a sufficiently narrow inferential chart. But a whole AGI mind, with multiple tasks, inconsistent data streams, shifting concepts, resource constraints, and evolving goals, should be represented as an atlas of such charts embedded in a paraconsistent evidence dynamics.

1.2 Contributions

This paper makes the following contributions.

1. It gives a compact technical review of νPLN and explains precisely which part of it is retained.
2. It reviews the historical PLN ideas most relevant to the present argument: contextual probability, multi-component truth values, weight-of-evidence, term logic, and inference control.

3. It uses p-bits and positive/negative evidence as the persistent truth-value substrate of πPLN.

4. It defines concrete quantales useful for πPLN, especially the evidence-count quantale

$$Q_{\pm}^{\text{cnt}} = ([0, \infty]^2, \leq, \vee, +, (0, 0)).$$

5. It proves that PLN STVs and the νPLN beta posterior truth values of [1] are projections of this quantale.

6. It gives a context-indexed revision calculus that does not require a global distribution: same-context empirical addition, overlap-aware discounting, corrective provenance operations, guarded cross-context translation, and residuated weakness-preserving revision.

7. It formulates πPLN as an atlas of local νPLN charts over contexts, without any global gluing axiom.

8. It shows how quantale weakness guides chart selection, generalization, inference control, and factor-graph scheduling.

9. It explains the practical difference for chainers and factor graphs: πPLN uses local Bayesian islands in a quantale-valued paraconsistent control graph.

2 A brief technical review of νPLN

This section reviews those parts of νPLN [1] needed for the present paper. The review is intentionally sympathetic: πPLN does not discard the local νPLN machinery. It reinterprets its scope.

2.1 The global probability space

The core νPLN construction begins with a probability space

$$(\Omega, \mathcal{F}, \nu),$$

where Ω is a set of possible worlds, $\mathcal{F}$ is a sigma algebra of events, and ν is a universal distribution. Each possible world contains a probabilistic predicate or generator $\widehat{\mu}$ and a complete history of its evaluations over a domain $\mathcal{D}$.

There is a random variable

$$M : \Omega \to \mathcal{L},$$

where $\mathcal{L}$ is a probabilistic programming language of possible predicate-generators, and a family of observable Boolean random variables

$$D_x : \Omega \to \{\text{False}, \text{True}\}, \qquad x \in \mathcal{D}.$$

For a finite observation set $S \subseteq \mathcal{D}$, the observed data are written D_S. The central Bayesian update is

$$\nu(M \in L \mid D_S) = \frac{\nu(M \in L)\nu(D_S \mid M \in L)}{\nu(D_S)},$$

or, suppressing M,

$$\nu(L \mid D_S) = \frac{\nu(L)\nu(D_S \mid L)}{\nu(D_S)}.$$

This is the heart of νPLN. Given a prior over possible models, update that prior using observed predicate evaluations.

2.2 Truth values as posterior distributions

In νPLN, a PLN truth value is naturally interpreted as a posterior distribution over a parameter or model class. For predicate direct introduction, the relevant model class is the Bernoulli family

$$\mathcal{B} = \{\text{Bernoulli}(\theta) : \theta \in [0, 1]\}.$$

Assume a beta prior

$$\theta \sim \text{Beta}(\alpha_0, \beta_0).$$

If N predicate observations contain n positive cases and $N - n$ negative cases, the posterior is

$$\theta \mid D_S \sim \text{Beta}(\alpha_0 + n, \beta_0 + N - n).$$

Equivalently, the cumulative distribution function is

$$F(x) = I_x(\alpha_0 + n, \beta_0 + N - n),$$

where I_x is the regularized incomplete beta function. This is a clean second-order-probability interpretation of direct introduction.

2.3 What νPLN gets right

νPLN gets several important things right:

1. It gives a rigorous local interpretation of PLN truth values as posterior distributions.
2. It clarifies how simple PLN strength-and-confidence values can be understood as compressed summaries of second-order distributions.
3. It provides a precise relationship between PLN and Solomonoff-style universal induction: both involve a second-order prior over possible generators and an update from finite observations.
4. It supplies a target semantics for certain PLN rule formulas: in an ideal local chart, the rule formulas should approximate Bayesian conditioning, marginalization, or posterior projection.

2.4 The scope error

The scope error is the claim that PLN as a whole should be defined by a single global distribution ν. For an embedded AGI system, this is too strong. It ignores at least five facts.

1. Evidence arrives in heterogeneous contexts that often do not share one natural sample space.
2. Concepts change their boundaries depending on task, perspective, and abstraction level.
3. A real system must carry mutually inconsistent evidence without explosion.

4. Inference is resource-bounded and task-relative, not passive prediction over one observation stream.
5. Independence and factorization assumptions are often useful local engineering choices, not global truths.

πPLN therefore preserves the local νPLN chart and rejects the global ν ontology.

3 Historical PLN: context, truth values, and inference control

This section recalls several historical PLN ideas that are essential for πPLN.

3.1 Term logic and probabilistic inheritance

PLN is grounded in a term-logic style of uncertain reasoning, in dialogue with NARS-style uncertain term logic [14]. The typical first-order relation is inheritance:

$$\text{Inheritance}(A, B),$$

which is interpreted extensionally as a probabilistic subset or conditional probability relation, and intensionally as a pattern/property-sharing relation. Predicate-side implication and concept-side inheritance are related by the usual concept/predicate isomorphism.

The practical virtue of term logic is that it gives simple schematic rules for deduction, induction, abduction, revision, and analogy. For example,

$$A \to B, \quad B \to C \quad \Rightarrow \quad A \to C.$$

In PLN this is accompanied by truth-value formulas estimating the strength and confidence of the conclusion from those of the premises.

3.2 Multi-component truth values

A PLN SimpleTruthValue is usually written

$$(s, c)$$

or

$$(s, n),$$

where $s \in [0, 1]$ is strength, $n \geq 0$ is weight-of-evidence or count, and c is a normalized confidence such as

$$c = \frac{n}{n + k}$$

for a system parameter $k > 0$.

Historical PLN also includes IndefiniteTruthValues and DistributionalTruthValues, reflecting both Bayesian/probabilistic motivations [10, 11] and concerns similar to imprecise-probability approaches [13]. The important point for the present paper is that PLN was never committed to a single scalar probability as its whole truth-value object. Second-order uncertainty and weight-of-evidence are fundamental.

3.3 Context

Historical PLN treats truth values as context-relative. A context can be understood as a local universe of discourse. In practice, the context may be universal, local, or somewhere between. The key idea is:

> PLN may construct local probability distributions based on local contexts rather than assuming one universal probability distribution for all inference.

This is precisely the instinct that πPLN formalizes.

3.4 Inference control

The PLN rule system tells us what inferences are possible. It does not by itself tell us which inference to perform next. Historical PLN already recognized inference control as central: forward and backward chaining generate large search trees that must be pruned by heuristics, attention, goals, uncertainty, and cost.

πPLN turns this into a formal principle. Inference control is not an afterthought; it is one of the main reasons the system needs contexts, weakness measures, provenance, and quantale-valued factor graphs.

4 Paraconsistent foundations

4.1 The p-bit idea

A p-bit is a paraconsistent evidence value with separate positive and negative components:

$$v = (v^+, v^-).$$

The four corners

$$(1, 0), \quad (0, 1), \quad (1, 1), \quad (0, 0)$$

represent, respectively, true-only, false-only, both true and false, and neither true nor false, in the broad Belnap/CD/paraconsistent family of ideas [8, 9, 2]. Interior points represent graded versions of these states.

The crucial point is that v^- is not $1 - v^+$. A statement may have high support and high opposition at the same time.

4.2 Positive and negative evidence

For πPLN, the more operational form is an evidence-count p-bit

$$(n^+, n^-) \in [0, \infty]^2.$$

Here n^+ is positive evidence and n^- is negative evidence. A prior-free or maximum-likelihood projection recovers the PLN strength and count by

$$s_{\text{emp}} = \frac{n^+}{n^+ + n^-}, \qquad n = n^+ + n^-.$$

Then confidence is obtained by

$$c = \frac{n}{n+k}.$$

This empirical projection is the $k = 0$ strength limit of the prior-aware projection introduced in Section 8.2. Thus the pair (n^+, n^-) contains the information needed for an STV (s, n) when $n > 0$, while also making the paraconsistent structure explicit before any prior is injected.

4.3 Contradiction as information

If both n^+ and n^- are large, the system should not collapse the statement to an uninformative average and forget the conflict. For example,

$$(n^+, n^-) = (1000, 1000)$$

has scalar strength 1/2, but it is very different from

$$(n^+, n^-) = (1, 1)$$

or from no evidence at all. The first indicates a major live contradiction; the second indicates a tiny amount of balanced evidence. A scalar STV projection with neutral prior deliberately loses much of this contradiction signal, retaining at most a confidence difference. This is exactly why the persistent metagraph stores raw positive and negative evidence rather than only projected beliefs. πPLN treats this difference as semantically important.

5 Quantale weakness and its role in logic

This section reviews the quantale-weakness perspective needed later.

5.1 Quantales

Definition 5.1 (Commutative quantale)**.** *A commutative quantale is a tuple*

$$Q = (Q, \leq, \bigvee, \otimes, e)$$

such that:

1. $(Q, \leq)$ *is a complete lattice with arbitrary joins* $\bigvee$.
2. $(Q, \otimes, e)$ *is a commutative monoid.*
3. *Tensor distributes over arbitrary joins:*

$$a \otimes \bigvee_i b_i = \bigvee_i (a \otimes b_i).$$

Intuitively, $\bigvee$ combines alternatives and $\otimes$ composes steps. This makes quantales natural for weighted logic, graph paths, fuzzy relations, resource costs, proof search, and factor graphs.

5.2 Weakness as generalized Occam

In set-theoretic form, a hypothesis is weak if it rules out few possibilities. If U is a universe of possible worlds and $H \subseteq U$ is the set of worlds where a hypothesis holds, then a simple weakness measure is

$$w(H) = |H|.$$

A larger satisfaction set means fewer exclusions, hence a weaker and more general hypothesis. This gives an Occam-like principle:

$$H^* = \arg\max_{H \supseteq D} w(H),$$

where D is the observed data.

Quantale weakness generalizes this idea. Instead of one cardinality measure over possible worlds, one chooses a quantale appropriate to the domain and defines weakness as an aggregate of low-distinction, low-effort, broadly applicable structure. In logic, weakness can be attached to propositions, proof paths, contexts, rules, or hypotheses. In the concrete πPLN architecture below, these carriers are specialized to charts, candidate revised states, proof steps, factors, and factor-graph messages. In factor graphs, weakness can be attached to factors and messages.

5.3 Weakness in logic

A local event algebra gives a simple logic quantale. Let $\mathcal{E}$ be a finite Boolean algebra of events, or more generally a complete Boolean algebra when arbitrary joins are needed. Then

$$Q_{\text{ev}} = (\mathcal{E}, \subseteq, \bigcup, \cap, \Omega)$$

is a commutative quantale: alternatives are unions, sequential logical combination is intersection, and the unit is the whole event space. The bottom is $\emptyset$, the empty join, but it is not an additional operation in the quantale signature.

For paraconsistent logic, one may use evidence-pair or CD-style quantales. The central idea is that proof and disproof are primitive, independently composable forms of justification. A statement's truth is not one number but a structured element of an ordered algebra.

5.4 Weakness and generalization

Weakness-guided induction says:

prefer the weakest hypothesis adequate to the evidence and task.

This is more general than Solomonoff or MDL. Solomonoff-style induction is recovered when the relevant weakness measure is essentially inverse program description length. But real agents use many other weakness notions: structural simplicity, contextual invariance, causal economy, low effort, low distinction-making, resonance across contexts, and transfer across task families.

6 Tasks, agency, and intelligence

For πPLN, inference is not passive prediction over one data stream. It is embedded in tasks, agency, and bounded resources.

Definition 6.1 (Task)**.** *A task is a tuple*

$$\tau = (\mathcal{O}, \mathcal{A}, \mathcal{G}, \mathcal{R}, H, S),$$

where $\mathcal{O}$ is an observation space, $\mathcal{A}$ an action space, $\mathcal{G}$ a goal or success structure, $\mathcal{R}$ a resource budget, H a horizon, and S an evaluation criterion.

A task family is a structured collection

$$\mathcal{T} = \{\tau_i\}_{i \in I}$$

with morphisms expressing transfer, abstraction, or refinement between tasks.

An inference chart is good not because it approximates a nonexistent global truth distribution, but because it helps the system succeed across relevant tasks and transfer to nearby tasks. Thus a chart is judged by fit, weakness, cost, provenance, conflict handling, and task value.

7 The basic architecture of πPLN

7.1 Persistent state

Let $\mathcal{L}$ be the system's logical or metagraph language, and let $\mathrm{Stmt}(\mathcal{L})$ be the set of statements expressible in it. The persistent knowledge state of πPLN is not a probability distribution. It is a context-indexed paraconsistent evidence metagraph whose fundamental storage unit is a provenance-tagged evidence packet.

For each statement-context pair, the metagraph manages a packet family

$$\mathcal{E}_t(\phi, C) = \{e_i = (\Delta_i^+, \Delta_i^-, P_i, a_i, r_i, \ldots)\}_i,$$

where P_i is provenance, a_i may include attention or temporal relevance, and r_i is a reliability or source-quality weight. The familiar count pair is a cached aggregate or projection of these packets:

$$(E^+_{\phi,C}, E^-_{\phi,C}) = \mathrm{Agg}_C(\mathcal{E}_t(\phi, C)) = \bigotimes_i r_i a_i (\Delta_i^+, \Delta_i^-),$$

with the tensor interpreted in the current evidence-count quantale and guarded by provenance-overlap policy.

Thus one may view the query interface as

$$K_t : \mathrm{Stmt}(\mathcal{L}) \times \mathrm{Ctx} \to \mathcal{P}(\mathrm{Packet}) \times [0, \infty]^2 \times \mathrm{Prov} \times \mathrm{Attn} \times \cdots .$$

For a statement ϕ in context C,

$$K_t(\phi, C) = (\mathcal{E}_t(\phi, C), (E^+_{\phi,C}, E^-_{\phi,C}), \mathrm{Prov}_{\phi,C}, \mathrm{Attn}_{\phi,C}, \ldots).$$

The packet family is the mathematically primary state. The aggregate pair is what ordinary PLN-style projections, local charts, and compatibility interfaces usually read. The remaining components record where the evidence came from, which assumptions and concept-boundary declarations were used, how much attention the evidence currently deserves, and what tasks it has helped or harmed. The same syntactic statement may therefore have multiple packet families and aggregate evidence states in different contexts, and these are not automatically merged. In a MeTTa/Hyperon implementation, a Space naturally stores the individual evaluated expressions or packet Atoms, while the aggregate truth object can be evaluated or cached on demand.

7.2 Contexts

Definition 7.1 (πPLN context)**.** *A* πPLN *context is a tuple*

$$C = (\mathcal{L}_C, X_C, E_C, T_C, A_C, W_C, \rho_C),$$

where:

1. $\mathcal{L}_C \subseteq \mathcal{L}$ *is a local language fragment.*
2. X_C *is a local universe of discourse.*
3. E_C *is selected evidence from* K_t*.*
4. T_C *is the current task, query, goal, or inference purpose.*
5. A_C *is a temporary assumption package.*
6. W_C *is a weakness structure or quantale.*
7. ρ_C *is a relevance/reliability/salience weight.*

The assumption package A_C may include independence assumptions, type restrictions, universe-size estimates, default priors, abstraction level, predicate boundary conventions, and temporary relevance filters.

7.3 Local νPLN charts

Definition 7.2 (Local chart constructor)**.** *A local chart constructor maps a persistent state and task-context to a temporary probabilistic chart:*

$$\Gamma_C(K_t, T_C) = (\Omega_C, \mathcal{F}_C, \nu_C, M_C, D^C, \iota_C).$$

Here $(\Omega_C, \mathcal{F}_C, \nu_C)$ *is a local probability space,* M_C *is a local model or predicate-generator random variable,* D^C *is a local observation family, and*

$$\iota_C : \mathrm{Stmt}(\mathcal{L}_C) \to \mathcal{F}_C$$

maps relevant statements to events or measurable predicates.

Inside C, one may do ordinary νPLN:

$$\nu_C(M_C \in L \mid D^C_S) = \frac{\nu_C(M_C \in L)\nu_C(D^C_S \mid M_C \in L)}{\nu_C(D^C_S)}.$$

The crucial negative axiom is:

$$\text{not required: } \exists(\Omega, \mathcal{F}, \nu)\ \forall C : \nu_C = \mathrm{Marg}_C(\nu).$$

Some contexts may glue. Others may not. Failure to glue is a datum, not a foundational catastrophe.

7.4 Projection back to evidence

The persistent metagraph stores empirical evidence, not prior-smeared belief. Thus the clean local-chart export is an evidence package

$$\mathsf{Out}_C(\phi) = (\widehat{n}_C^+(\phi), \widehat{n}_C^-(\phi), \mathrm{Prov}_C, A_C),$$

where A_C records the assumptions, priors, predicate boundaries, and factorizations used by the chart. If a legacy interface exports a local empirical STV

$$\mathrm{TV}_C^{\mathrm{emp}}(\phi) = (s_C, n_C),$$

where s_C is the empirical fraction and n_C the empirical count, then πPLN converts this into evidence increments

$$\Delta_C^+(\phi) = \rho_C n_C s_C,$$
$$\Delta_C^-(\phi) = \rho_C n_C (1 - s_C).$$

If instead the local chart exports a Bayesian STV whose strength already contains a context prior $(p_{0,C}, k_C)$, then the prior must be unwound before persistent storage. Writing $N_C = n_C$ for the empirical count represented by the confidence component, the empirical increments are

$$\Delta_C^+(\phi) = \rho_C \left((N_C + k_C)s_C - k_C p_{0,C} \right),$$
$$\Delta_C^-(\phi) = \rho_C \left(N_C - ((N_C + k_C)s_C - k_C p_{0,C}) \right),$$

with clipping or diagnostic failure if the result falls outside $[0, N_C]$. This makes explicit the architectural rule: priors belong to contexts and projections; the durable metagraph stores evidence.

Crucially, these increments are not dumped into a universal bucket for ϕ. To prevent ontology smear, where a strict concept boundary in one context is indiscriminately averaged with a loose concept boundary in another, the persistent update is context-indexed:

$$K_{t+1}(\phi, C) = K_t(\phi, C) \otimes (\Delta_C^+(\phi), \Delta_C^-(\phi), \mathrm{Prov}_C).$$

Evidence aggregation using $\otimes$ occurs automatically only within the same context, or across subcontexts explicitly marked as ontologically identical by the current context-management policy. If a process needs to evaluate ϕ across mutually incompatible contexts C_A and C_B, it does not merge their counts. It constructs a covering decision context and explicit translation maps, or else records the discrepancy as a higher-order paraconsistent property of ϕ.

A covering decision context C_* for contexts $C_1, \ldots, C_m$ is a temporary context containing a decision-specific language fragment, assumption package, and translation maps $\tau_{i \to *} : C_i \to C_*$ for just the concepts needed by the current task. It is not a universal context. It is a local chart-selection and ontology-alignment artifact whose purpose is to make a particular cross-context projection well typed.

In the simplest evidence-count model, $\otimes$ is addition of positive and negative counts, with provenance-aware correction for overlapping evidence.

8 Concrete quantales for πPLN

This section answers the technical question: what quantale should πPLN use? The answer is not a single magic quantale. πPLN uses a quantale interface, with several important concrete instantiations.

8.1 The evidence-count quantale

Definition 8.1 (Evidence-count quantale). *Let*

$$Q_{\pm}^{\mathrm{cnt}} = ([0,\infty]^2, \leq, \vee, \otimes, e),$$

where

$$(n^+, n^-) \leq (m^+, m^-) \quad \textit{iff} \quad n^+ \leq m^+ \textit{ and } n^- \leq m^-,$$
$$(n^+, n^-) \vee (m^+, m^-) = (\max(n^+, m^+), \max(n^-, m^-)),$$
$$(n^+, n^-) \otimes (m^+, m^-) = (n^+ + m^+, n^- + m^-),$$

and

$$e = (0, 0).$$

This is the primary truth/evidence quantale of πPLN.

Conceptually, $\vee$ takes the least upper bound of alternative evidence bounds, while $\otimes$ composes independent or provenance-disjoint evidence by adding counts.

Theorem 8.2 (Evidence-count quantale). *$Q_{\pm}^{\mathrm{cnt}}$ is a commutative quantale.*

The proof is in Appendix B.

8.2 Projection to PLN STVs

The evidence-count pair (n^+, n^-) represents empirical, paraconsistent data. It does not itself contain a prior. Priors are properties of contexts, assumption packages, or projections from evidence into decision-time beliefs.

Let $p_0 \in [0,1]$ be the prior expected strength for the relevant context, and let $k > 0$ be the prior weight or virtual count. Define

$$\Pi_{\mathrm{STV}}^{p_0,k}(n^+, n^-) = (s, c),$$

where

$$s = \frac{n^+ + kp_0}{n^+ + n^- + k},$$
$$c = \frac{n^+ + n^-}{n^+ + n^- + k}.$$

If $n^+ + n^- = 0$, this gives $s = p_0$ and $c = 0$. As empirical evidence grows, s approaches the empirical mean $n^+/(n^+ + n^-)$ and c approaches 1. Because $Q_{\pm}^{\mathrm{cnt}}$ allows extended counts in $[0,\infty]^2$, STV projections involving infinite counts are evaluated by limits along finite approximants. For example, if $n^+ \to \infty$ while n^- and k remain finite, then $s \to 1$ and $c \to 1$; similarly, if $n^- \to \infty$ while n^+ remains finite, then $s \to 0$ and $c \to 1$. This avoids treating expressions such as infinite empirical counts as literal undefined divisions in implementation. The prior-free empirical projection is recovered as the limiting or diagnostic case $k = 0$:

$$s_{\mathrm{emp}} = \frac{n^+}{n^+ + n^-}.$$

Thus the metagraph stores raw evidence counts, while the chart or decision context constructs the belief projection.

8.3 Projection to beta posteriors

The beta distribution gives the direct bridge to νPLN predicate direct introduction [1]. Choose beta prior parameters to match the STV prior:

$$\alpha_0 = kp_0, \qquad \beta_0 = k(1 - p_0).$$

Define

$$\Pi_{\text{Beta}}(n^+, n^-) = \text{Beta}(\alpha_0 + n^+, \beta_0 + n^-).$$

The mean of this beta distribution is

$$\frac{\alpha_0 + n^+}{\alpha_0 + \beta_0 + n^+ + n^-} = \frac{n^+ + kp_0}{n^+ + n^- + k},$$

which is exactly the STV strength in Section 8.2.

Theorem 8.3 (Local beta recovery)**.** *Let C be a πPLN context whose local chart satisfies the Bernoulli assumptions of predicate direct introduction. Let D_S^C contain n^+ positive and n^- negative empirical observations of a predicate ϕ, and assume a beta prior derived from the context's assumption package A_C,* $\text{Beta}(\alpha_0, \beta_0)$*. Then*

$$\Pi_{\text{Beta}}(n^+, n^-) = \text{Beta}(\alpha_0 + n^+, \beta_0 + n^-)$$

is exactly the posterior beta truth distribution derived in the local νPLN calculation [1].

The key insight is that (n^+, n^-) is a sufficient statistic for the local Bernoulli chart, while (α_0, β_0) belongs to the local chart's prior rather than to the persistent evidence object.

8.4 Why evidence-count tensor is Bayesian likelihood multiplication

For a Bernoulli parameter θ, the evidence count (n^+, n^-) corresponds to the likelihood

$$L_{(n^+,n^-)}(\theta) = \theta^{n^+}(1 - \theta)^{n^-}.$$

Combining independent bodies of evidence gives

$$L_{(n_1^+,n_1^-)}(\theta)L_{(n_2^+,n_2^-)}(\theta) = \theta^{n_1^+ + n_2^+}(1 - \theta)^{n_1^- + n_2^-}.$$

Thus the tensor

$$(n_1^+, n_1^-) \otimes (n_2^+, n_2^-) = (n_1^+ + n_2^+, n_1^- + n_2^-)$$

is exactly Bayesian likelihood multiplication written in sufficient-statistic coordinates.

8.5 The bounded p-bit propagation quantale

For fast propagation, resonance, and attention control, bounded values are often more convenient than unbounded counts. The bounded quantale used for evidence accumulation should be a genuine reflection of the count quantale, not a diluting reliability calculus.

Definition 8.4 (Bounded p-bit propagation quantale)**.** *Let*

$$Q_{\pm}^{[0,1]} = ([0,1]^2, \leq, \vee, \otimes, e),$$

where order and join are componentwise. Define the tensor by componentwise probabilistic sum, also called the algebraic sum t-conorm:

$$(a^+, a^-) \otimes (b^+, b^-) = (a^+ + b^+ - a^+b^+,\ a^- + b^- - a^-b^-),$$

and define the unit by

$$e = (0, 0).$$

This quantale is useful for activation flow, resonance propagation, bounded factor-graph message passing, and attention control where one wants strong independent bounded evidence to accumulate rather than dilute. The older componentwise-multiplication p-bit quantale remains useful for path reliability or preservation-through-a-chain readings, but it is not the correct bounded image of evidence-count addition.

A monotone change-of-base map from raw counts to bounded values is

$$\rho_\lambda(n^+, n^-) = (1 - e^{-n^+/\lambda}, 1 - e^{-n^-/\lambda}).$$

With the probabilistic-sum tensor, ρ_λ is a strict homomorphism from count accumulation to bounded accumulation:

$$\rho_\lambda(n_1^+, n_1^-) \otimes \rho_\lambda(n_2^+, n_2^-) = \rho_\lambda(n_1^+ + n_2^+,\ n_1^- + n_2^-).$$

Indeed, for one coordinate, if $a = 1 - e^{-n_1/\lambda}$ and $b = 1 - e^{-n_2/\lambda}$, then

$$1 - e^{-(n_1+n_2)/\lambda} = 1 - (1-a)(1-b) = a + b - ab.$$

This loses raw count information but preserves the algebraic structure of accumulation. Propagating bounded activations through the metagraph therefore mirrors the bounded projection of accumulated evidence rather than artificially attenuating it.

Theorem 8.5 (Bounded accumulation quantale and change of base)**.** $Q_{\pm}^{[0,1]}$ *with probabilistic-sum tensor is a commutative quantale, and* ρ_λ *is a strict quantale homomorphism from* $Q_{\pm}^{\mathrm{cnt}}$ *to* $Q_{\pm}^{[0,1]}$*: it preserves the monoidal tensor and, because* $1 - e^{-x/\lambda}$ *preserves arbitrary suprema on* $[0, \infty]$*, it also preserves componentwise joins.*

8.6 Evidence order versus truth order

There are two important orders on p-bit-like values.

The evidence or knowledge order is

$$(n^+, n^-) \leq_K (m^+, m^-) \quad \text{iff} \quad n^+ \leq m^+,\ n^- \leq m^-.$$

More support and more opposition both mean more information.

The truth or logical order is

$$(x, x') \leq_T (y, y') \quad \text{iff} \quad x \leq y \text{ and } y' \leq x'.$$

More positive evidence makes a statement truer; more negative evidence makes it less true. The evidence-count quantale uses the evidence order. CD-style logic uses the truth order.

8.7 A CD logic quantale

Let H be a complete Boolean algebra, or more generally a complete bi-Heyting algebra with the needed distributivity. Define

$$Q_{\mathrm{CD}} = H \times H^{op}.$$

Under the truth order,

$$(x, x') \leq_T (y, y') \quad \text{iff} \quad x \leq y \text{ and } y' \leq x',$$

we may define

$$(x, x') \vee_T (y, y') = (x \vee y, x' \wedge y'),$$

$$(x, x') \otimes_T (y, y') = (x \wedge y, x' \vee y'),$$

with unit

$$e = (1, 0).$$

This is a useful logical quantale for paraconsistent proof and disproof. However, for operational PLN evidence storage, $Q_{\pm}^{\mathrm{cnt}}$ is usually the better primitive.

8.8 The local event quantale

Inside a finite local chart C, let $\mathcal{E}_C$ be the Boolean algebra of events generated by the local variables. Finiteness makes it a complete Boolean algebra; for infinite charts one should explicitly use a complete event algebra rather than a merely finite Boolean algebra or merely a sigma-algebra. Then

$$Q_C^{\mathrm{ev}} = (\mathcal{E}_C, \subseteq, \bigcup, \cap, \Omega_C)$$

is a commutative quantale, with $\emptyset$ as the bottom element.

The local probability valuation

$$\nu_C : \mathcal{E}_C \to [0, 1]$$

is not generally a quantale homomorphism from intersection to multiplication, because

$$\nu_C(A \cap B) = \nu_C(A)\nu_C(B)$$

holds only under independence. Therefore probability maps should be treated as context-local, assumption-dependent, and generally lax.

This is one of the precise mathematical points where πPLN differs from global νPLN: factorization assumptions are local chart data, not global semantic truths.

Accordingly, probability logic and evidence logic are structurally separated. The event quantale Q_C^{ev} and the probability valuation ν_C are used inside the local Bayesian island, where dependencies are explicitly modeled by the chart. The global control graph does not pass events or probabilities from Q_C^{ev}. It passes exported evidence elements in $Q_{\pm}^{\mathrm{cnt}}$ or labels in $Q_{\pi\mathrm{PLN}}$, where the quantale laws are strict.

8.9 The product control quantale

For practical inference control, an inference step should carry more than a truth value. It should carry evidence, assumptions, weakness, cost, and provenance.

Let

$$Q_{\pi\mathrm{PLN}} = Q_{\mathrm{assump}} \times Q_{\pm}^{\mathrm{cnt}} \times Q_{\mathrm{weak}} \times Q_{\mathrm{cost}} \times Q_{\mathrm{prov}}.$$

A concrete choice is:

$$Q_{\mathrm{assump}} = (\mathcal{P}(\mathcal{A}), \subseteq, \cup, \cup, \emptyset),$$

tracking assumptions used;

$$Q_{\pm}^{\mathrm{cnt}} = ([0,\infty]^2, \leq, \vee, +, (0,0)),$$

tracking positive and negative evidence;

$$Q_{\mathrm{weak}} = ([0,1], \leq, \max, \times, 1),$$

tracking normalized weakness, reliability, or generality under composition;

$$Q_{\mathrm{cost}} = ([0,\infty], \geq, \inf, +, 0),$$

tracking computational cost in tropical style, where lower cost is better;

$$Q_{\mathrm{prov}} = (\mathcal{P}(\mathrm{Sources}), \subseteq, \cup, \cup, \emptyset),$$

tracking evidence sources.

The product of commutative quantales is a commutative quantale. Thus each inference action a may be labeled by

$$\ell(a) = (A_a, (n_a^+, n_a^-), w_a, c_a, P_a) \in Q_{\pi\mathrm{PLN}}.$$

Sequential composition uses $\otimes$ componentwise. Alternative derivations use $\vee$ componentwise. Scalar priorities used by implementation are projections of this richer label, not the semantics itself.

9 Formal definition of πPLN

9.1 The base language

Let $\mathcal{L}$ be a typed metagraph language with statements $\mathrm{Stmt}(\mathcal{L})$ and inference rules

$$r : (\phi_1, \ldots, \phi_k) \Rightarrow \psi.$$

A statement may be first-order, higher-order, extensional, intensional, temporal, causal, or meta-level. The formalism below does not depend on a specific choice of PLN relation types.

9.2 Persistent evidence semantics

Definition 9.1 (πPLN knowledge state)**.** *A* πPLN *knowledge state at time t is a context-indexed metagraph mapping whose fundamental values are provenance-tagged evidence packets:*

$$K_t : \mathrm{Stmt}(\mathcal{L}) \times \mathrm{Ctx} \to \mathcal{P}(\mathrm{Packet}) \times Q_{\pm}^{\mathrm{cnt}} \times \mathcal{M}.$$

Here Ctx *is the space of system contexts and $\mathcal{M}$ is metadata including provenance, attentional values, task traces, reliability estimates, assumption fingerprints, and ontology-boundary declarations. The $Q_{\pm}^{\mathrm{cnt}}$ component is the cached aggregate projection of the packet family, not the primitive storage object.*

If we query a statement ϕ inside a specific context C, we retrieve

$$K_t(\phi, C) = (\mathcal{E}_t(\phi, C), ((E^+_{\phi,C}, E^-_{\phi,C}), m_{\phi,C})).$$

Here

$$(E^+_{\phi,C}, E^-_{\phi,C}) = \mathrm{Agg}_C(\mathcal{E}_t(\phi, C))$$

is the aggregate count projection used by ordinary truth-value formulas. The packet set $\mathcal{E}_t(\phi, C)$ remains available for corrective revision, provenance-sensitive discounting, contradiction diagnosis, and ontology relabeling.

Scalar projections are optional and context-specific. With prior strength $p_{0,C}$ and virtual count k_C, define

$$s_t(\phi, C) = \frac{E^+_{\phi,C} + k_C p_{0,C}}{E^+_{\phi,C} + E^-_{\phi,C} + k_C},$$

$$c_t(\phi, C) = \frac{E^+_{\phi,C} + E^-_{\phi,C}}{E^+_{\phi,C} + E^-_{\phi,C} + k_C}.$$

These are projections for decision, communication, or local chart construction. They are not the base semantics.

If the system requires a scalar assessment across multiple contexts, it must actively construct a covering decision context C_* and translation maps

$$\tau_{i\to*} : C_i \to C_*.$$

Only after these maps certify that the relevant concept boundaries and assumptions are compatible may the system aggregate the pulled-back or pushed-forward evidence. Without such active translation, the persistent state remains a localized, multi-world evidence store.

9.3 Local chart semantics

Definition 9.2 (Charted inference episode)**.** *A charted inference episode is a tuple*

$$\mathcal{I} = (K_t, C, \Gamma_C, \Phi_C, R_C, B_C),$$

where K_t is the current knowledge state, C is a context, Γ_C is a local chart, Φ_C is a set of target statements, R_C is a selected rule set, and B_C is a resource budget.

Inside Γ_C, ordinary probabilistic reasoning may be used. The output is a set of local truth values

$$\mathrm{TV}_C(\phi)$$

for $\phi \in \Phi_C$. These truth values may be STVs, intervals, distributions, or factor-graph marginals. They are projected back into evidence-count form.

9.4 No-global-distribution principle

Principle 9.3 (No global probability space)**.** πPLN *does not assume there exists a single probability space* $(\Omega, \mathcal{F}, \nu)$ *such that every local chart* $(\Omega_C, \mathcal{F}_C, \nu_C)$ *is a marginal, conditioning, or restriction of it.*

When such a global distribution exists for a family of contexts, it may be constructed and used. But gluing is an achievement, not an axiom.

9.5 The atlas view

Let $\mathcal{C}$ be a category or preorder of contexts. πPLN defines a partial, lax chart assignment

$$\mathsf{Chart} : \mathcal{C}^{op} \dashrightarrow \mathsf{ProbChart}.$$

For each context C in the domain,

$$C \mapsto (\Omega_C, \mathcal{F}_C, \nu_C, M_C, D^C, \iota_C).$$

For overlapping contexts C, D, there may be a partial translation

$$\tau_{CD} : \mathcal{L}_C \supseteq \mathcal{L}_{CD} \to \mathcal{L}_D.$$

These context morphisms are the formal carriers of the guarded translations used later in cross-context revision and covering decision contexts. Local chart outputs may disagree. The discrepancy

$$\delta_{CD}(\phi) = d(\mathrm{TV}_C(\phi), \mathrm{TV}_D(\tau_{CD}\phi))$$

is recorded as conflict, not erased.

10 Relation to νPLN

10.1 Local conservativity

Theorem 10.1 (Local νPLN conservativity)**.** *Let C be a context and suppose all statements, observations, priors, and factorization assumptions used in a derivation lie inside the chart* $\Gamma_C = (\Omega_C, \mathcal{F}_C, \nu_C, M_C, D^C, \iota_C)$*. Then the local probabilistic calculations of* πPLN *restricted to C are exactly the corresponding* νPLN *calculations in that chart.*

This theorem is conceptually central. πPLN is not anti-Bayesian. It is anti-global-Bayesian as a foundational doctrine for AGI.

10.2 Non-gluability

A family of local probability charts need not admit a global joint probability distribution, even when all pairwise overlaps agree. This is the probabilistic, not merely logical, obstruction that matters for a chart-atlas semantics: a non-extendible-marginals phenomenon in the Vorob'ev/Bell-style family of examples [7].

Theorem 10.2 (Non-extendible pairwise marginals)**.** *Let X, Y, Z be $\{-1, +1\}$-valued variables. For each pair, define the same locally coherent unbiased pairwise marginal:*

$$P(X = +1, Y = +1) = P(X = -1, Y = -1) = \frac{1}{16},$$

$$P(X = +1, Y = -1) = P(X = -1, Y = +1) = \frac{7}{16},$$

and analogously for (Y, Z) and (X, Z). Each two-variable chart is a valid probability distribution and all one-variable overlaps are uniform. However, there is no global probability distribution on (X, Y, Z) having these three pairwise marginals.

Each local chart says that its two variables are strongly anti-correlated, with correlation

$$r = -\frac{3}{4}.$$

The numerical choice $r = -3/4$ is intentionally past the positive-semidefinite extendibility boundary $r = -1/2$ for three equicorrelated variables. If a global joint distribution existed, the correlation matrix

$$R = \begin{pmatrix} 1 & r & r \\ r & 1 & r \\ r & r & 1 \end{pmatrix}$$

would have to be positive semidefinite. Its eigenvalues are $1 - r, 1 - r, 1 + 2r$. For $r = -3/4$, the last eigenvalue is $1 + 2r = -1/2 < 0$, impossible. Thus the local charts are genuinely probabilistic and overlap-compatible, but non-extendible.

The earlier Boolean triangle

$$A = B, \quad B = C, \quad A \neq C$$

is still useful as the simplest logical cartoon of non-gluability, but the unbiased anti-correlation example is the stronger riposte to a global probabilistic gluing assumption.

10.3 Exact gluing as a special case

When local charts are projectively consistent over a suitable inverse system of finite event algebras, they can be glued. πPLN accepts this. It simply refuses to make it foundational.

Principle 10.3 (Gluing as an achievement)**.** *If a family of local charts satisfies the compatibility conditions needed for a joint chart, use the joint chart. If it does not, retain the local charts and record the obstruction as paraconsistent evidence.*

11 Revision in πPLN without a global distribution

Revision is the most serious objection to a non-global semantics. If one assumes that every judgment in the system is a marginal or conditional from one master probability distribution, then belief revision has a familiar ideal form: condition the global distribution on the new evidence. The natural worry may be phrased as follows: without a global distribution, what does it mean to revise properly rather than merely pile up inconsistent numbers?

The answer is that πPLN changes the target. Revision is not defined as conditioning one universal probability distribution. Revision is defined as context-indexed evidence transformation. In a local Bayesian chart it may reduce to ordinary conditioning, beta-binomial updating, or factor-graph message passing. In the persistent metagraph it is a more general operation on empirical p-bit/count evidence with provenance, contexts, ontology tags, temporal relevance, and weakness constraints.

The guiding slogan is:

Bayesian conditioning is one projection of revision, not the definition of revision.

11.1 Historical PLN revision and the need for more structure

Historical PLN already recognized that revision cannot be done sensibly with a single scalar probability. Different estimates of an Atom's truth value may come from external sources, internal cognitive processes, or different pathways of PLN reasoning. The usual simple truth-value revision rule therefore uses both strength and evidence count. In its simplest form, if two estimates are (s_1, n_1) and (s_2, n_2), then

$$s = \frac{n_1 s_1 + n_2 s_2}{n_1 + n_2}.$$

If the bodies of evidence overlap, PLN discounts the resulting count, for example by a heuristic of the form

$$n = n_1 + n_2 - c \min(n_1, n_2),$$

where $c = 0$ means independence and $c = 1$ means complete redundancy. Historical PLN also recognizes temporal revision, inference trails, and the possibility that evidence previously accepted may later be found fallacious. These are exactly the phenomena that a global probability ideal tends to hide: they concern provenance, overlap, time, and correction of the evidence base itself.

πPLN preserves the old PLN insight but gives it a cleaner algebraic home. The persistent store contains not just

$$\phi \mapsto (s, c),$$

but

$$K_t(\phi, C) = ((E^+_{\phi,C}, E^-_{\phi,C}), m_{\phi,C}),$$

where $m_{\phi,C}$ contains provenance, source reliability, attention, task traces, temporal tags, context links, and ontology/translation data. Revision acts on these richer objects.

11.2 Raw evidence versus projected belief

A central distinction is between raw empirical evidence and projected belief. The evidence-count pair

$$(n^+, n^-)$$

means: positive and negative empirical support recorded for a statement in a context. It does not already include a beta prior. Priors enter only when a local chart or decision procedure asks for a scalar or distributional projection:

$$(n^+, n^-) \mapsto \mathrm{Beta}(\alpha_0 + n^+, \beta_0 + n^-),$$

or

$$(n^+, n^-) \mapsto \left(\frac{n^+ + kp_0}{n^+ + n^- + k}, \frac{n^+ + n^-}{n^+ + n^- + k} \right).$$

Thus two contexts may inspect the same empirical evidence but project it using different priors, default assumptions, or universe-size conventions. The metagraph stores facts about evidence; the chart constructs the belief.

This is the precise sense in which πPLN can bypass the need for a prior at the persistent-revision level. It does not claim to produce a Bayesian posterior without a prior. Rather, it says that revision of the empirical metagraph can happen before Bayesian projection.

11.3 Same-context empirical revision

The simplest case is same-context empirical revision. Suppose a new evidence packet e concerns the same statement ϕ in the same context C and has provenance disjoint from the currently stored evidence. Let

$$\Delta(e) = (\Delta^+(e), \Delta^-(e)) \in Q_{\pm}^{\mathrm{cnt}}.$$

Then

$$K_{t+1}(\phi, C) = K_t(\phi, C) \otimes \Delta(e),$$

meaning, in the count quantale,

$$(E^+_{\phi,C}, E^-_{\phi,C}) \otimes (\Delta^+, \Delta^-) = (E^+_{\phi,C} + \Delta^+, E^-_{\phi,C} + \Delta^-).$$

This is prior-free empirical addition. If the system later projects to an STV with no explicit prior, the resulting strength is the usual weighted empirical mean. If it projects with a prior, the same raw counts become a posterior mean.

11.4 Overlap-aware revision

Disjointness is rarely guaranteed. Multiple proof paths may share premises; two documents may quote the same source; two sensors may be coupled; a chainer may rederive a conclusion from a trail that already included it. Naive addition would inflate confidence.

Let $P_t(\phi, C)$ be the set or weighted family of provenance tokens already supporting the evidence for (ϕ, C), and let $P(e)$ be the provenance of the new packet. Define an *independence coefficient*, also called a novelty coefficient,

$$\lambda^{\pm}(e; K_t) \in [0, 1],$$

where 1 means fully new evidence and 0 means fully redundant evidence. A simple overlap-aware count update is

$$E^+_{t+1} = E^+_t + \lambda^+(e; K_t)\Delta^+(e),$$
$$E^-_{t+1} = E^-_t + \lambda^-(e; K_t)\Delta^-(e).$$

The coefficients may be computed from explicit source overlap, inferred statistical dependence, proof-trail intersection, shared document lineage, common sensor channel, or a local factor-graph dependence estimate. Calling λ an independence or novelty coefficient avoids a terminology inversion: an overlap coefficient would naturally have the opposite convention, with 1 meaning fully overlapped.

More abstractly, one may replace ordinary tensor by a guarded tensor

$$\otimes_{\text{Prov}} : Q_{\pm}^{\text{cnt}} \times Q_{\pm}^{\text{cnt}} \times \mathcal{P}(\text{Prov})^2 \to Q_{\pm}^{\text{cnt}},$$

whose behavior is ordinary addition on disjoint provenance and discounted addition on overlapping provenance. This is the πPLN version of the old PLN count correction

$$n = n_1 + n_2 - c \min(n_1, n_2),$$

but now applied at the level of context-indexed evidence packets rather than collapsed truth values.

11.5 Corrective revision

Sometimes new information does not merely oppose an old conclusion; it says that some old evidence should not have been counted as evidence of that kind at all. A source may be discovered fraudulent, a sensor may have been miscalibrated, a document may be duplicated, a concept boundary may have been misapplied, or an old observation may be obsolete.

For this case, πPLN stores evidence in provenance-indexed packets

$$\mathcal{E}_t(\phi, C) = \{e_i = (\Delta_i^+, \Delta_i^-, P_i, a_i, r_i, \ldots)\}_i.$$

The collapsed count is a projection:

$$(E_{\phi,C}^+, E_{\phi,C}^-) = \bigotimes_i r_i a_i (\Delta_i^+, \Delta_i^-),$$

where r_i is reliability and a_i may include temporal relevance or attention. Corrective revision acts on packets before projection:

$$\mathcal{E}_{t+1}(\phi, C) = \text{Corr}(\mathcal{E}_t(\phi, C), P_0, \eta),$$

where P_0 identifies suspect provenance and η specifies the correction. Examples include:

1. *Quarantine:* retain the packet but mark it unusable for ordinary aggregation until reviewed.
2. *Discount:* lower its reliability r_i.
3. *Retraction:* remove it from the active projection.
4. *Relabeling:* move it from context C to a more accurate context C' or from statement ϕ to translated statement ϕ'.
5. *Counterevidence generation:* add evidence for a meta-statement such as "source S is unreliable" without simply pretending that the original observation was evidence for $\neg\phi$.

This distinguishes two situations that scalar Bayesian updating tends to blur:

$$\text{new evidence against } \phi$$

versus

$$\text{new evidence that old support for } \phi \text{ was invalidly counted.}$$

Only the former should normally increase $E_{\phi,C}^-$. The latter should modify, quarantine, or relabel provenance.

11.6 Temporal and relevance revision

Historical PLN suggests that temporal discounting may be handled by cognitive processes outside the pure logical formula. πPLN makes this explicit by placing time and relevance in metadata. If e_i was observed at time t_i and used at time t, define a temporal relevance weight

$$a_i(t) = d_C(t - t_i, \mathrm{src}(e_i), \phi, T),$$

where d_C is context- and task-sensitive. For a stable physical constant, d_C may decay hardly at all; for a person's current location, it may decay rapidly. The active evidence projection becomes

$$(E^+_{\phi,C}(t), E^-_{\phi,C}(t)) = \bigotimes_i r_i a_i(t)(\Delta_i^+, \Delta_i^-).$$

Revision can therefore request a temporal update before combining two packets, just as classical PLN recommends requesting an update of stale premises before revision, deduction, induction, or abduction.

11.7 Cross-context guarded revision

The persistent metagraph is context-indexed to avoid ontology smear. Evidence for ϕ in context D is not automatically evidence for ϕ in context C. Cross-context revision requires an explicit translation map

$$\tau_{D\to C} : \mathrm{Stmt}(\mathcal{L}_D) \dashrightarrow \mathrm{Stmt}(\mathcal{L}_C)$$

and a translation quality or reliability

$$r_{D\to C}(\psi) \in Q_{\pi\mathrm{PLN}}.$$

If $\tau_{D\to C}(\psi) = \phi$, then an admissible transported update has the form

$$K_{t+1}(\phi, C) = K_t(\phi, C) \otimes_{\mathrm{Prov}} (r_{D\to C}(\psi) \otimes K_t(\psi, D)).$$

If no adequate translation exists, the evidence remains separate:

$$K_t(\phi, C), \qquad K_t(\psi, D),$$

and their discrepancy is represented as a higher-order conflict or incommensurability statement rather than collapsed into one bucket.

For example, a strict medical context and a loose commonsense context may both use the string "fever." πPLN does not average their evidence merely because the symbol matches. It requires a translation such as

$$\tau(\mathrm{Fever}_{\mathrm{commonsense}}(x)) = \mathrm{Temperature}(x) > 37.8^\circ\mathrm{C}$$

with explicit reliability and scope.

11.8 Residuated weakness revision

The more general revision rule uses residuation. Let

$$Q_C = (Q_C, \leq_C, \bigvee_C, \otimes_C, e_C, \Rightarrow_C)$$

be a commutative residuated quantale attached to context C. The residual is characterized by

$$a \otimes_C b \leq_C c \quad \iff \quad b \leq_C (a \Rightarrow_C c).$$

It is the algebraic "gap" or "difference" operator. In probability-like quantales it behaves like a ratio or conditional mass; in logic quantales it is implication; in cost quantales it is remaining budget or slack; in p-bit/product quantales it is computed componentwise or channelwise.

Let H be the current local hypothesis, relation, rule set, proof agenda, or chart family, and let E be new evidence. Let

$$\mathcal{K}_C(H, E)$$

be the set of candidate revised states generated locally by adding/deleting links, revising a proof agenda, mutating a program, splitting a concept, changing an independence assumption, or selecting a nearby chart.

The key typing issue is that a candidate state and an evidence packet are not always elements of the same set. A proof agenda, a program patch, a chart family, and a sensor packet do not literally intersect. To compute residual fit and preservation without a category mistake, πPLN projects all these objects into a shared logical lattice inside the context.

Let $\mathcal{P}_C$ be the complete lattice of evaluable propositions, patterns, or local metagraph assertions in context C, and let

$$w_C : \mathcal{P}_C \to Q_C$$

be the context's weakness valuation. Let

$$\mathrm{eval}_{\mathcal{K}} : \mathcal{K}_C(H, E) \longrightarrow \mathcal{P}_C$$

map a candidate state or hypothesis into its logical extension in $\mathcal{P}_C$, and let

$$\mathrm{eval}_{\mathcal{E}} : \mathcal{E}_C \longrightarrow \mathcal{P}_C$$

map an evidence packet into the same lattice. The native lattice meet $\wedge$ then represents explicit pattern-matching intersection, coverage, or common extension inside the context space.

Define residual fit and residual preservation by

$$\mathrm{fit}_C(K; E) = w_C(\mathrm{eval}_{\mathcal{E}}(E)) \Rightarrow_C w_C(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E)),$$

$$\mathrm{pres}_C(K; H) = w_C(\mathrm{eval}_{\mathcal{K}}(H)) \Rightarrow_C w_C(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{K}}(H)).$$

In typed implementations, the projection maps pull model and data down to Atoms within a unified context space, often a MeTTa Space or related metagraph subspace, where $\wedge$ is an actual pattern-matching meet or pullback. This is the algebraic lattice-pullback form of conservative revision.

For hard evidence with threshold τ, define

$$\mathrm{Rev}_{C,\tau}(H, E) = \underset{\substack{K \in \mathcal{K}_C(H,E) \\ \mathrm{fit}_C(K;E) \geq_C \tau}}{\arg\overset{\mathrm{lex}}{\max}} \left(\mathrm{pres}_C(K; H), w_C(\mathrm{eval}_{\mathcal{K}}(K))\right).$$

For soft evidence, define

$$\mathrm{Rev}_C(H, E) = \underset{K \in \mathcal{K}_C(H,E)}{\arg\overset{\mathrm{lex}}{\max}} \left(\mathrm{fit}_C(K; E), \mathrm{pres}_C(K; H), w_C(\mathrm{eval}_{\mathcal{K}}(K))\right).$$

This is the πPLN conservative revision principle:

> First accommodate the new evidence; then preserve as much useful old structure as possible; then prefer the weakest, least-committal adequate revision.

11.9 Probability-valued special case

Let $Q = ([0,1], \leq, \times, 1, \sup, \Rightarrow)$, where

$$a \Rightarrow b = \begin{cases} 1, & a \leq b, \\ b/a, & 0 < b < a, \\ 0, & b = 0 < a, \end{cases}$$

with the usual harmless convention at $a = 0$. If $\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E) \leq \mathrm{eval}_{\mathcal{E}}(E)$ in $\mathcal{P}_C$, then

$$\mathrm{fit}(K; E) = w(\mathrm{eval}_{\mathcal{E}}(E)) \Rightarrow w(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E)) = \frac{w(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E))}{w(\mathrm{eval}_{\mathcal{E}}(E))}.$$

So residual fit is exactly the local conditional mass of the revised state's coverage inside the evidence. Thus Bayesian-looking revision is recovered when the valuation is probabilistic. But the formula only needs the local mass of $\mathrm{eval}_{\mathcal{E}}(E)$ and $\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E)$ in the current chart or context. It does not require one global probability distribution over all possible cognitive experience.

11.10 Evidence-count and beta-binomial special case

For Bernoulli direct evidence in one context, a state is summarized by

$$H = (n^+, n^-).$$

New independent evidence is

$$E = (m^+, m^-).$$

Same-context empirical revision gives

$$\mathrm{Rev}(H, E) = (n^+ + m^+, n^- + m^-).$$

Projection to a beta posterior gives

$$\mathrm{Beta}(\alpha_0 + n^+ + m^+, \beta_0 + n^- + m^-),$$

which is the same posterior one obtains by first projecting H and then performing a local beta-binomial Bayesian update. Therefore local νPLN beta revision is recovered as a projection of empirical count revision.

This theorem is the clean reply to the worry that no proper revision is possible without a global distribution. Proper beta revision requires a local prior, but empirical count revision does not. The prior enters the chart; it does not live in the persistent evidence tensor.

11.11 Valuation revision

Sometimes the evidence does not primarily require changing H; it requires changing the valuation that decides what counts as weak, salient, relevant, or credible. Let $\{\mu_\theta\}_{\theta \in \Theta}$ be candidate valuations, each inducing $w_{C,\theta}$, $\mathrm{fit}_{C,\theta}$, and $\mathrm{pres}_{C,\theta}$. Then valuation revision chooses

$$(\theta^*, K^*) = \underset{\theta \in \Theta,\ K \in \mathcal{K}_{C,\theta}(H,E)}{\overset{\mathrm{lex}}{\arg\max}} \left(\mathrm{fit}_{C,\theta}(K; E), \mathrm{pres}_{C,\theta}(K; H), w_{C,\theta}(K), -\mathrm{Cost}(\theta) \right).$$

This is the formal counterpart of learning that one's old salience function, independence assumption, concept boundary, or habit prior was inappropriate. It replaces "assume the right prior" with "revise the local valuation that plays the prior-like role."

11.12 Layer and ontology revision

If no candidate K in the current context can fit the evidence above threshold, the system may move to a different abstraction layer or ontology. Let

$$F : C \to C'$$

be a context morphism, translation, enrichment, or refinement. Layer revision chooses

$$(C', K') = \underset{C', K}{\arg\overset{\text{lex}}{\max}} \left(-\mathrm{Cost}(C, C'), \mathrm{fit}_{C'}(K; E), \mathrm{pres}_{C'}(K; F(H)), w_{C'}(K)\right).$$

This captures concept splitting, concept merging, moving from Boolean to paraconsistent truth values, moving from symbolic to vector features, adding an unmodeled causal variable, or replacing a defective local chart by a richer one.

Layer revision is especially important for avoiding ontology smear. Instead of forcing incompatible concept boundaries to average, πPLN can introduce a new covering context, keep the original local evidence intact, and explicitly store the translations and residual conflicts.

11.13 Revision as inference control

Revision is not a rare cleanup operation. It is a first-class inference-control move. A chainer action may be

$$a = (\text{revise}, H, E, C, \mathcal{K}_C, \text{budget}),$$

scored by expected fit gain, preservation loss, weakness, conflict resolution, and cost. A factor graph may include revision factors:

$$F_{\mathrm{Rev}}(H, E, K) = (\mathrm{fit}_C(K; E), \mathrm{pres}_C(K; H), w_C(K), \mathrm{Cost}(K)),$$

whose messages are quantale-valued evidence/control summaries rather than global probability marginals.

This is where the formalism matters practically. A νPLN chainer tempted by a global distribution asks for the updated posterior. A πPLN chainer asks which local revision move gives the best tradeoff among evidence fit, old-structure preservation, weakness, provenance safety, ontology safety, and compute.

11.14 Summary of the reply to the global-revision objection

If revision means globally coherent Bayesian conditioning over all beliefs, then a global distribution is required. But this is too narrow for an AGI metagraph. In πPLN:

1. Persistent revision acts on empirical p-bit/count evidence with context and provenance.
2. Local Bayesian posterior revision is recovered when a context constructs a Bayesian chart.
3. Priors are local projection parameters, not global foundations.
4. Overlap, time, correction, and ontology translation are explicit revision operations.
5. Residuated weakness gives a general conservative revision rule: fit new evidence, preserve old structure, and select the weakest adequate revision.

12 Recovering PLN rules inside πPLN

12.1 Rule lifting

Let

$$r : (\phi_1, \ldots, \phi_k) \Rightarrow \psi$$

be a PLN rule with a local truth-value function

$$f_r^C : \mathrm{TV}_C(\phi_1) \times \cdots \times \mathrm{TV}_C(\phi_k) \to \mathrm{TV}_C(\psi).$$

The πPLN lift is:

$$\widehat{f_r} : K_t(\phi_1, C), \ldots, K_t(\phi_k, C), C \to \Delta_C(\psi),$$

where $\Delta_C(\psi)$ is a positive/negative evidence contribution with context and provenance.
The lifted rule has three stages:

$$\text{select/build local chart} \quad \to \quad \text{compute local PLN truth value} \quad \to \quad \text{project to p-bit evidence.}$$

12.2 Revision and provenance

Section 11 gave the general revision calculus. At the level of rule lifting, the key point is that every lifted rule emits a packet of evidence rather than merely a final scalar truth value. If two evidence contributions are independent or provenance-disjoint, simple addition is admissible:

$$(n_1^+, n_1^-) \otimes (n_2^+, n_2^-) = (n_1^+ + n_2^+, n_1^- + n_2^-).$$

If their provenance overlaps, naive addition double-counts evidence. In that case, revision requires an overlap correction. A simple form is

$$(n_1, n_2) \mapsto n_1 + n_2 - \lambda \, \mathrm{overlap}(n_1, n_2),$$

applied componentwise or at the level of likelihood factors. The exact correction is implementation-dependent, but the requirement is semantic: evidence aggregation must preserve provenance.

In πPLN, this subsection is not a mere engineering footnote. The provenance-aware revision step is part of the semantics of rule application. A rule conclusion is stored as

$$(\psi, C, \Delta^+, \Delta^-, \mathrm{Prov}_r, A_C, T_C),$$

and can later be added, discounted, corrected, quarantined, translated, or reprojected according to the revision rules of Section 11. This is how πPLN avoids pretending that different proof trails are independent samples from one global distribution.

12.3 Implication direct introduction

For an implication

$$P \Rightarrow Q,$$

the local Bernoulli direct-introduction chart restricts attention to cases where P is true. Positive evidence counts cases where P and Q are both true; negative evidence counts cases where P is true and Q is false:

$$n^+ = |\{x : P(x) \wedge Q(x)\}|,$$
$$n^- = |\{x : P(x) \wedge \neg Q(x)\}|.$$

Then

$$\mathrm{TV}_{\pi,C}(P \Rightarrow Q) = (n^+, n^-)$$

projects to

$$\mathrm{Beta}(\alpha_0 + n^+, \beta_0 + n^-)$$

or to the corresponding STV.

12.4 Conjunction direct introduction

The νPLN draft [1] also spends substantial effort on conjunction introduction. The πPLN treatment is deliberately local and evidence-count based. Inside a single chart C where A and B are jointly observable on a common sample set S_C, define

$$n^+_{A\wedge B,C} = |\{i \in S_C : A_i = 1,\ B_i = 1\}|,$$

$$n^-_{A\wedge B,C} = |\{i \in S_C : \neg(A_i = 1 \text{ and } B_i = 1)\}|.$$

Then the conjunction truth object is the empirical count pair

$$TV_{\pi,C}(A \wedge B) = (n^+_{A\wedge B,C}, n^-_{A\wedge B,C}),$$

which can be projected to an STV or beta posterior exactly as in Sections 8.2 and 8.3.

For a tiny worked example, suppose S_C has ten jointly observed cases and $A \wedge B$ is true in cases $1, 2, 3$ only. Then

$$(n^+_{A\wedge B,C}, n^-_{A\wedge B,C}) = (3, 7).$$

With beta prior parameters $\alpha_0 = \beta_0 = 1$, the local chart projection is

$$\mathrm{Beta}(1 + 3, 1 + 7) = \mathrm{Beta}(4, 8),$$

with posterior mean $4/12 = 1/3$. This is exactly the usual Bernoulli/binomial posterior for the local event "A and B are jointly true" in a chart where joint observations are available. Continuous or combinatorial conjunction derivations of the νPLN style therefore live inside such a joint chart; πPLN adds only the rule that the joint chart must be explicit and context-local.

For implication-style direct introduction $A \Rightarrow B$, the negative count is usually chosen differently:

$$n^+_{A\Rightarrow B,C} = |\{i \in S_C : A_i = 1,\ B_i = 1\}|,$$

$$n^-_{A\Rightarrow B,C} = |\{i \in S_C : A_i = 1,\ B_i = 0\}|,$$

with cases $A_i = 0$ ignored or used for other rule estimates. The point is not that one universal convention handles all connectives; the point is that the counting convention belongs to the local chart and rule schema.

If A is observed in context C_A and B in context C_B, there is no well-typed conjunction count until the system constructs a joint chart C_{AB} or an explicit translation/covering context with a common sample set. This is where conjunction introduction touches the central πPLN issue: ordinary probabilistic formulas for conjunction are valid inside a chart with a specified joint observation space, but they should not be silently applied across incommensurable contexts. A global ν formulation can represent the joint space if its model class and prior already contain the relevant factorization; πPLN makes the construction of that joint space an explicit inference-control act.

12.5 Indefinite and distributional truth values

An IndefiniteTruthValue may be represented as a set or distribution of possible evidence-count pairs:

$$\mathcal{I}_\phi \subseteq [0,\infty]^2$$

or

$$\eta_\phi \in \mathcal{P}([0,\infty]^2).$$

A DistributionalTruthValue may similarly be treated as a distribution over strengths, over beta parameters, or over evidence-count pairs. πPLN prefers the last representation because it preserves paraconsistent structure.

13 Weakness-guided chart selection

13.1 The problem

Given a task T and a target statement ϕ, there are often many possible contexts and charts:

$$C_1, \ldots, C_m.$$

They may differ in abstraction level, universe size, prior, factorization, predicate boundary, independence assumptions, cost, and relevance. Which chart should be used?

A global ν answer says: use the chart dictated by the global probability space. πPLN says: choose charts by a task-relative weakness principle.

13.2 Weakest adequate chart

Let H range over candidate hypotheses, models, or charts inside context C. Let $w_C(H)$ be the weakness of H in the quantale Q_C. Let $\mathrm{Fit}_C(H, E_C)$ measure fit to selected evidence. Let $\mathrm{Rel}_T(H)$ measure task relevance, and $\mathrm{Cost}(H)$ computational cost.

A constrained selection principle is

$$H_C^* = \arg\max_H w_C(H)$$

subject to

$$\mathrm{Fit}_C(H, E_C) \geq \theta,$$

$$\mathrm{Rel}_T(H) \geq \rho,$$

$$\text{Cost}(H) \leq B.$$

Equivalently, one may use a combined score

$$\text{Score}_C(H) = \text{Fit}_C(H, E_C) \otimes \text{Weak}_C(H) \otimes \text{Rel}_T(H) \otimes \text{Cost}(H)^{-1}.$$

The operations here are interpreted in the relevant product quantale, with scalarization only for scheduling.

13.3 MDL and Solomonoff-style priors as special cases

If the weakness quantale is chosen so that

$$w(H) \propto 2^{-K(H)},$$

where $K(H)$ is program description length, then weakness-guided chart selection recovers the MAP/MDL side of a Solomonoff-style universal prior [15, 16]. Strict Solomonoff induction is a mixture over all programs, not merely an $\arg\max$ over one program. Thus the result here is not that πPLN literally reduces to Solomonoff induction, but that the usual description-length or universal-prior preference is one specialization of quantale weakness. Other tasks may require other weakness measures: causal weakness, conceptual weakness, graph-structural weakness, p-bit conflict weakness, or task-transfer weakness.

13.4 Why this matters for real agents

A real agent is not merely predicting the next bit of an observation stream. It is choosing actions, allocating attention, forming concepts, revising goals, and transferring competence across tasks. Therefore induction must be indexed by tasks and contexts. Weakness is the general meta-bias; Solomonoff-style program priors and MDL are important realizations, but not the whole story.

14 Practical inference control: chainers

14.1 The πPLN chainer state

A πPLN chainer state is not merely an Atomspace plus a rule library. It is

$$S_t = (G_t, K_t, \mathcal{A}_t, Q_t, T_t, B_t, R_t),$$

where G_t is the metagraph, K_t the paraconsistent evidence state, $\mathcal{A}_t$ the active chart atlas, Q_t the agenda, T_t the current task or query, B_t the resource budget, and R_t the rule library.

A candidate inference step is

$$a = (r, \text{premises}, \text{target}, C, \Gamma_C, \text{cost}, \text{prov}).$$

It carries a quantale label

$$\ell(a) = (A_a, (n_a^+, n_a^-), w_a, c_a, P_a) \in Q_{\pi\text{PLN}}.$$

14.2 Composition and alternatives

Sequential proof composition uses tensor:

$$\ell(a_1 \circ a_2) = \ell(a_1) \otimes \ell(a_2).$$

Alternative proof routes use join:

$$\ell(a_1 \vee a_2) = \ell(a_1) \vee \ell(a_2).$$

A priority queue may use a scalar projection, for example

$$\text{Priority}(a) = \lambda_1 U_T(a) + \lambda_2 w_a + \lambda_3 \Delta\text{conf}(a) - \lambda_4 \text{ConflictPenalty}(a) - \lambda_5 \text{Cost}(a),$$

but this scalar priority is an implementation policy, not the semantics.

14.3 Revision moves as first-class chainer actions

A πPLN chainer should treat revision as an ordinary candidate action. A revision action has the form

$$a_{\text{Rev}} = (\text{Rev}, H, E, C, \mathcal{K}_C, B),$$

where H is the current local proof agenda, hypothesis, or evidence state; E is the new evidence or contradiction; C is the active context; $\mathcal{K}_C$ is the local candidate generator; and B is the budget. The action is scored by

$$\text{Score}(a_{\text{Rev}}) = F(\text{fit}_C, \text{pres}_C, w_C, \text{Cost}, \text{conflict}, \text{task utility}),$$

where fit_C and pres_C are the residual fit and preservation terms from Section 11. This lets a chainer decide, for instance, whether to add counterevidence, quarantine a suspect proof path, split a context, or simply leave conflicting evidence in place until the task requires a projection.

A backward chainer uses revision when a newly discovered proof path overlaps with an old one or contradicts it. A forward chainer uses revision when a background inference changes the support/opposition balance of an Atom or reveals that some old source should be discounted. Neither operation requires constructing a global posterior over the whole Atomspace.

14.4 Backward chaining

For a target ϕ, a πPLN backward chainer:

1. selects candidate contexts $C_1, \ldots, C_m$;
2. constructs or retrieves charts Γ_{C_i};
3. expands proof trees using ordinary PLN rules inside each chart;
4. evaluates local truth values;
5. projects local results to p-bit/count evidence for ϕ;
6. schedules further expansion by weakness, expected evidence gain, conflict relevance, task utility, and cost.

14.5 Forward chaining

For background inference, a πPLN forward chainer:

1. detects changes in K_t or G_t;
2. identifies affected context neighborhoods;
3. constructs small local charts around hot spots;
4. propagates p-bit evidence;
5. records new statements with provenance and context tags;
6. promotes only statements whose task relevance, weakness, novelty, or conflict value justifies attention.

14.6 Why this differs from νPLN

In a global νPLN interpretation, the ideal chainer approximates the best global posterior update. In πPLN, the chainer manages an ecology of local charts. It must decide not just which rule to apply, but which temporary world to construct for the rule, and how to store the result without pretending that all other temporary worlds agree.

15 Practical inference control: factor graphs

15.1 Quantale-valued factor graphs

Let $G = (V, F, E)$ be a factor graph, with variables V, factors F, and incidence relation E. In ordinary probabilistic factor graphs, messages are numbers or probability vectors. In πPLN, messages may be elements of a commutative quantale Q.

For variable-to-factor messages:

$$m_{v\to f}(x_v) = \bigotimes_{g\in N(v)\setminus\{f\}} m_{g\to v}(x_v).$$

For factor-to-variable messages:

$$m_{f\to v}(x_v) = \bigvee_{x_{N(f)\setminus\{v\}}} \left[\phi_f(x_{N(f)}) \otimes \bigotimes_{u\in N(f)\setminus\{v\}} m_{u\to f}(x_u) \right].$$

This is ordinary sum-product generalized by replacing sum with join and product with tensor. The exactness claim for tree-shaped graphs concerns quantale evidence marginals, not global probability marginals. Local probabilities and their dependencies are handled inside chart-specific factors; once exported, the global graph routes strict evidence/control-quantale labels.

15.2 Local Bayesian islands

A subgraph corresponding to a local chart C may use ordinary probabilistic sum-product internally. It computes local marginals or sufficient statistics. The result is then exported as an evidence-count element

$$(n_C^+, n_C^-) \in Q_\pm^{\mathrm{cnt}}.$$

Thus the architecture is:

$$\begin{gathered}\text{local Bayesian factor graph inside } C\\ \Downarrow \\ \text{p-bit/count evidence export} \\ \Downarrow \\ \text{global quantale-valued control graph.}\end{gathered}$$

The whole graph is not one global joint probability distribution. It is a quantale-valued control and aggregation structure coordinating local Bayesian islands.

Revision factors obey the same quarantine. If a local Bayesian island revises a beta posterior, the result exported to the global graph is an evidence-count increment, a provenance correction, or a translated p-bit packet. The global quantale graph never needs to pass around the island's entire posterior as if it were part of one universal probability model.

15.3 Factor types

A practical πPLN factor graph may include:

1. PLN rule factors f_r;
2. chart-likelihood factors f_Γ;
3. context-compatibility factors f_C;
4. weakness factors f_W;
5. provenance-overlap factors f_P;
6. revision factors f_{Rev} carrying residual fit, preservation, weakness, and correction/quarantine choices;
7. task-utility factors f_T;
8. conflict-diagnostic factors f_{conflict}.

Here f_{conflict} is implementation-defined but must be monotone in both support and opposition. A simple count-level trigger used in examples is

$$\min(n^+, n^-) \geq \theta_{\min} \quad \text{and} \quad n^+ + n^- \geq \theta_N,$$

possibly with provenance-separation requirements. More refined versions may return a quantale value measuring balanced high evidence, source independence, and task relevance.

This gives a single graph architecture for background inference, local Bayesian updating, rule scheduling, attention allocation, and conflict monitoring.

16 Examples

16.1 Birds, penguins, and flying: a worked example

This subsection works the canonical default-with-exceptions example through πPLN with explicit numbers, in order to make the paraconsistency payoff concrete rather than rhetorical. The setup is deliberately simple: one target predicate, two named individuals, and a handful of contexts. Even at this scale, the (n^+, n^-) representation preserves at least four distinct facts that scalar or globally-merged probabilistic representations conflate.

Setup: the persistent state

Suppose the system has accumulated evidence about flying behaviour from three sources:

- A field study of 1000 birds in a temperate aviary, of which 950 were observed to fly and 50 were not. This evidence is recorded under C_{bird}, a context whose assumption package $A_{C_{\text{bird}}}$ declares "bird" to mean "ordinary non-flightless bird" and whose universe $X_{C_{\text{bird}}}$ explicitly excludes penguins, ostriches, kiwis and similar taxa.
- An Antarctic study of 100 penguins, none of which were observed to fly. This evidence is recorded under C_{penguin}, whose assumption package treats Penguin as a species-level concept and inherits no flying default from the bird context.
- A zoological registration record asserting Bird(Tweety) and Penguin(Tweety) with full positive evidence in a species-classification context C_{zoo}.

The persistent metagraph K_t then contains:

$$K_t(\text{Fly}(x), C_{\text{bird}}) = ((950, 50), \text{Prov}_1, A_{C_{\text{bird}}}, \ldots),$$
$$K_t(\text{Fly}(x), C_{\text{penguin}}) = ((0, 100), \text{Prov}_2, A_{C_{\text{penguin}}}, \ldots),$$
$$K_t(\text{Bird}(\text{Tweety}), C_{\text{zoo}}) = ((1, 0), \text{Prov}_3, \ldots), \qquad K_t(\text{Penguin}(\text{Tweety}), C_{\text{zoo}}) = ((1, 0), \text{Prov}_3, \ldots).$$

The crucial structural fact is that the two flying counts live in different contexts, with different concept-boundary assumptions, and they are not automatically combined.

Query 1: does Tweety fly?

The system constructs a temporary decision context C_* for the query Fly(Tweety). Both C_{bird} and C_{penguin} are candidate sources of evidence. Two facts about the chart-selection step matter:

1. C_{penguin} is the more specific context: its declared universe is a strict subclass of the universe declared by C_{bird}.[1]
2. C_{bird}'s assumption package explicitly excludes penguins. Therefore its evidence does not apply to Tweety at all; using it would be a misuse of the chart.

[1] Note that this is *less weak* in the hypothesis-class sense of Section 10, but more relevant in the context-applicability sense. The two notions of weakness are distinct, and a practical chart selector trades them off.

So C_* is built from C_{penguin}'s evidence only. Project to an STV with a generic "birds usually fly" prior $p_0 = 0.95, k = 2$:

$$s = \frac{0 + 2 \cdot 0.95}{0 + 100 + 2} = \frac{1.9}{102} \approx 0.0186, \qquad c = \frac{100}{102} \approx 0.98.$$

The system concludes that Tweety almost certainly does not fly, with high confidence. Equivalently, Π_{Beta} gives a posterior $\text{Beta}(1.9 + 0,\ 0.1 + 100) = \text{Beta}(1.9, 100.1)$, sharply peaked near zero.

The key point is not the numerical answer. It is that this answer was obtained *without ever combining penguin evidence with the bulk bird evidence*. The 950 flying birds did not contribute to a global "probability of flight given bird" which would then be conditioned on Penguin. They contributed to a different chart with a different scope.

Query 2: does Robin fly?

Robin is a sparrow. Bird(Robin) holds in C_{zoo} and Penguin(Robin) does not. Now C_{bird} is the relevant chart, and its assumption package is satisfied (Robin is exactly the kind of bird the temperate-aviary study sampled). The projection from $(950, 50)$ under the same prior gives

$$s = \frac{950 + 1.9}{1000 + 2} \approx 0.950, \qquad c = \frac{1000}{1002} \approx 0.998.$$

The same persistent state therefore supports two opposite confident predictions, one for Tweety and one for Robin, with no contradiction at the level of the metagraph, because the contradictory-looking evidence lives in different contexts and is applied through different charts.

The naive-merge pathology

Now consider what happens if a system were to insist on one global rate of flying-given-bird. Merging counts across C_{bird} and C_{penguin}:

$$(950, 50) \otimes (0, 100) = (950, 150), \qquad s_{\text{emp}} = \frac{950}{1100} \approx 0.864.$$

This number is sample-mix-dependent, not concept-dependent. Had the Antarctic team observed ten times as many penguins:

$$(950, 50) \otimes (0, 1000) = (950, 1050), \qquad s_{\text{emp}} = \frac{950}{2000} \approx 0.475.$$

Same underlying biology; the "global probability that a bird flies" has moved from 0.86 to 0.48 purely because of where field crews went. In a νPLN-style global account, this is recoverable in principle by conditioning on a finer-grained predicate (Bird $\wedge$ $\neg$Penguin versus Bird $\wedge$ Penguin), but only if the model class $\mathcal{L}$ contains such hierarchical predicates and the prior over $\mathcal{L}$ is set up to recover the right factorisation from the observed data. In πPLN the partitioning is provided up front, by the assumption packages of the two contexts. The 0.864 number is never computed because there is no chart in which it is meaningful.

Ontology smear: incompatible concept boundaries

The story so far concerned two contexts whose universes happen to be disjoint. A more diagnostic case arises when two contexts *use the same word differently*. Suppose two ornithology groups have reported observations of flying:

- Group A operationalises "Bird" permissively (includes penguins, ostriches, kiwis, dead specimens scored as "did not fly"). Their study yields $(n_A^+, n_A^-) = (60, 40)$ in a context C_A.
- Group B operationalises "Bird" strictly (only volant taxa in the breeding season). Their study yields $(n_B^+, n_B^-) = (95, 5)$ in a context C_B.

These contexts share the syntactic predicate symbol Bird but their assumption packages disagree about its referent. The persistent metagraph keeps them separate:

$$K_t(\mathrm{Fly}(x), C_A) = (60, 40), \qquad K_t(\mathrm{Fly}(x), C_B) = (95, 5).$$

A merger without translation produces

$$(60, 40) \otimes (95, 5) = (155, 45), \qquad s_{\mathrm{emp}} \approx 0.775,$$

which is meaningless: it averages two incompatible operationalisations into a single number that neither group would endorse. πPLN blocks this by construction: aggregation across C_A and C_B requires an explicit translation map $\tau_{AB} : \mathcal{L}_A \supseteq \mathcal{L}_{AB} \to \mathcal{L}_B$ certifying that the relevant predicate boundaries are compatible. If the system tries to answer "does this bird fly?" without first picking a definition of "bird," the chart selector refuses, or returns a context-tagged disjunction. This is the structural prophylaxis against ontology smear that the πPLN context-indexed update law is designed to provide.

Live contradiction in a single context

The most explicitly paraconsistent case arises when two studies share a context but disagree. Suppose within C_{bird} two independent field surveys both target the same operationally defined population of non-penguin birds, and report:

- Survey 1 (high-altitude meadows): $(480, 20)$ from 500 birds.
- Survey 2 (urban downtown): $(50, 450)$ from 500 birds.

Both surveys claim to sample the same concept. Their results are radically different. If the evidence is naively tensored:

$$(480, 20) \otimes (50, 450) = (530, 470).$$

Using the prior-free empirical strength projection gives

$$s_{\mathrm{emp}} = \frac{530}{1000} \approx 0.530.$$

Using the ordinary PLN confidence map with system confidence parameter $k = 2$ gives

$$c_{k=2} = \frac{1000}{1002} \approx 0.998.$$

The scalar report "a near-certain 53% flying rate" is therefore a mixed projection: empirical strength plus high confidence from large total count. It is misleading in a strong sense. Neither survey supports a 53% rate; both support near-deterministic answers in opposite directions. The pair $(530, 470)$ does *contain* the relevant information — both components are large, which is the signature of a live contradiction — but the STV projection erases it. This is the precise sense in which the "$(1000, 1000)$ versus $(1, 1)$" point from Section 4.3 is operational: high-and-balanced evidence flags conflict, low-and-balanced evidence flags ignorance, and an STV strength of 0.5 cannot distinguish them.

πPLN responds by preserving the two surveys as separate provenance-tagged evidence entries within C_{bird}, and by exposing a *conflict-diagnostic factor* f_{conflict} (Section 15.3) that fires when both n^+ and n^- are simultaneously large relative to a context-specific threshold. Concretely, the system may store

$$K_t(\text{Fly}(x), C_{\text{bird}}) = ((480, 20), \text{Prov}_{\text{meadow}}) \oplus ((50, 450), \text{Prov}_{\text{urban}}),$$

where $\oplus$ is provenance-preserving aggregation (not the count tensor) and the high-and-balanced signature $(\min(n^+, n^-) \geq 50) \wedge (n^+ + n^- \geq 100)$ triggers a non-trivial value of f_{conflict}. The chainer then treats this as a goal: the live contradiction is something to investigate, not something to average away. A typical resolution is to refine C_{bird} into $C_{\text{bird,meadow}}$ and $C_{\text{bird,urban}}$ — in effect, learning that the operational "Bird" of C_{bird} was concealing two subpopulations.

This refinement step is itself a paraconsistent move: nothing was retracted, nothing was averaged, and the original evidence remains available with full counts in case the apparent dichotomy turns out to be spurious.

Tweety as a paraconsistent atom

A subtler payoff appears when one tracks evidence about a particular individual. Suppose Tweety, the penguin, is once filmed gliding briefly off a high ledge. Optimistic observers report this as a positive instance of Fly(Tweety). Sceptical observers classify it as falling. Each side logs evidence in its own context:

$$K_t(\text{Fly}(\text{Tweety}), C_{\text{optimist}}) = (1, 0), \qquad K_t(\text{Fly}(\text{Tweety}), C_{\text{sceptic}}) = (0, 1).$$

Meanwhile the bird-context default chart, applied through the Tweety-is-a-bird classification, can be used to project a positive contribution; and the penguin-context default chart projects a negative contribution. The persistent state for the bare statement Fly(Tweety) acquires the structure

$$\begin{aligned} K_t(\text{Fly}(\text{Tweety}), \cdot) = \{&((\Delta^+_{C_{\text{bird}}}, 0), C_{\text{bird}}, A_{C_{\text{bird}}}), \\ &((0, \Delta^-_{C_{\text{penguin}}}), C_{\text{penguin}}, A_{C_{\text{penguin}}}), \\ &((1, 0), C_{\text{optimist}}), \\ &((0, 1), C_{\text{sceptic}})\}. \end{aligned}$$

There is no classical-logical step from these entries to an arbitrary conclusion: no rule in the πPLN system lifts "positive evidence in one context, negative evidence in another" to $\perp$ and thence to anything else. The Atom for Fly(Tweety) is, in Belnap's terminology, in state Both, and the system can perfectly well continue to reason about Tweety without first resolving its flight status. A decision context specialised to ornithological adjudication may project these four entries to a scalar via a chosen reliability weighting on the contexts; a decision context specialised to airworthiness clearly favours the sceptical chart; neither projection retroactively rewrites the persistent state.

Revision without a global distribution: the mistaken-default case

The example also shows how revision proceeds without a global probability space. Suppose a backward chainer mistakenly applies the ordinary-bird default to Tweety and emits a low-confidence positive packet

$$e = (\text{Fly}(\text{Tweety}), C_{\text{penguin}}, (\Delta^+, \Delta^-), \text{Prov}_e) = ((9, 1), \text{Prov}_e)$$

inside the penguin context. The existing local state is

$$H : K_t(\text{Fly}(x), C_{\text{penguin}}) = (0, 100).$$

A scalar revision formula would be tempted to average these into a slightly less certain anti-flight conclusion, or into a muddled compromise. πPLN instead treats e as a revision problem over candidate local states.

Let $\mathcal{P}_{C_{\text{penguin}}}$ be the proposition lattice generated by local statements such as

$$\text{Penguin}(x), \quad \text{Fly}(x), \quad \text{VideoReliable}(e), \quad \text{AidedGlide}(x), \quad \text{CartoonOrToy}(x).$$

Candidate revisions include:

1. K_0: simply add the positive packet to the penguin count;
2. K_1: quarantine the packet as a bad default inference from the wrong context;
3. K_2: revise the classification and infer that Tweety is not actually a penguin;
4. K_3: preserve the penguin default but add an exception subcontext, $C_{\text{penguin,aided-glide}}$;
5. K_4: split the media context from the biological context, e.g. real penguins versus cartoon, toy, or glider-aided cases.

For each candidate K_i, compute

$$\text{fit}_C(K_i; e) = w_C(\text{eval}_{\mathcal{E}}(e)) \Rightarrow_C w_C(\text{eval}_{\mathcal{K}}(K_i) \wedge \text{eval}_{\mathcal{E}}(e)),$$

$$\text{pres}_C(K_i; H) = w_C(\text{eval}_{\mathcal{K}}(H)) \Rightarrow_C w_C(\text{eval}_{\mathcal{K}}(K_i) \wedge \text{eval}_{\mathcal{K}}(H)).$$

Here is a concrete toy calculation using the probability-like residual from Section 11. Let $\mathcal{P}_C$ be a finite local proposition lattice whose universe has 100 equally weighted microcases. Let $w_C(P) = |P|/100$, so weaker propositions cover more of the local universe. The new packet e has extension

$$\text{eval}_{\mathcal{E}}(e) = E, \qquad w_C(E) = 0.10,$$

representing the 10 positive-looking video frames. Suppose the old penguin non-flight default has extension H with $w_C(H) = 0.95$. Consider three candidate revisions:

	$w_C(\text{eval}_{\mathcal{K}}(K_i) \wedge E)$	$\text{fit}_C(K_i; e)$	$\text{pres}_C(K_i; H)$
K_0 add packet blindly	0.10	1.00	0.80
K_1 quarantine as wrong-context default	0.02	0.20	1.00
K_3 add aided-glide exception subcontext	0.09	0.90	0.98

where, for example,

$$\text{fit}_C(K_3; e) = 0.10 \Rightarrow 0.09 = 0.09/0.10 = 0.90$$

in the probability residual. If the task threshold is $\tau = 0.85$, then K_1 fails fit, while K_0 and K_3 both fit. Lexicographic selection by preservation then prefers K_3 over K_0 because $0.98 > 0.80$, while also retaining high fit. Thus the selected revision is not a compromise probability; it is a local structural change: preserve the penguin default, but add an exception or media subcontext explaining this packet.

The naive addition K_0 has high fit but poor preservation, because it weakens a high-value local default without explaining why this packet is exceptional. The reclassification K_2 may have high fit if the zoological record is weak, but poor preservation if the species record is strong. The exception or media-split revisions K_3, K_4 usually have the best conservative profile: they fit the video packet while preserving the useful generalization "penguins do not normally fly." The revision selected depends on task and weakness valuation, but it is computed locally over the penguin and media context lattices, not by conditioning a global distribution over all possible animal worlds.

Revision of ontology smear

The two-group operationalisation case also calls for revision. Suppose a later calibration packet e_{cal} reveals that Group A counted dead specimens and ostriches as birds, while Group B counted only living volant taxa. The evidence packet does not revise the global proposition "birds fly." It revises the translation relation

$$\tau_{A\to B} : \mathrm{Bird}_A(x) \dashrightarrow \mathrm{Bird}_B(x).$$

Candidates include:

1. strengthen $\tau_{A\to B}$ and merge the counts anyway;
2. weaken $\tau_{A\to B}$ and keep the counts separate;
3. introduce an intermediate covering context with explicit subtypes VolantBird, FlightlessBird, and Specimen;
4. relabel Group A's old evidence into the covering context and preserve Group B as a stricter subcontext.

The residuated rule scores these candidates by fit to the calibration packet, preservation of the original studies, and weakness of the induced ontology. A well-designed weakness measure typically prefers the intermediate covering context: it explains why the two old rates differed, avoids averaging incompatible counts, and exports a usable translation map for future queries. Again, no global distribution is needed; the revision target is the context atlas itself.

What is preserved and what is lost

Table 1 summarises the contrast between three representations of the same evidence body. Each row corresponds to a piece of cognitive content that an AGI plausibly needs to act on; each column to a representation strategy.

The structural pattern in Table 1 is consistent. Where the single-distribution columns require additional model machinery to recover a distinction — richer model class, finer factorisation, observer-noise parameters, hierarchical priors — πPLN inherits the distinction from the evidence-storage layer. In each case the additional machinery in a νPLN-style account is doing the work that πPLN delegates to context-indexing and to keeping n^+ and n^- separate.

Cognitive content	Scalar STV	Single global ν	πPLN (n^+, n^-) atlas
"Most birds fly" as a generalisation	yes, as $s \approx 0.95$	yes, as a posterior	yes, in C_{bird}
"Penguins do not fly" as a generalisation	only via a separate STV	only with hierarchical model class	yes, in C_{penguin}
Sample-mix-independent subclass rates	no	only with the right model class	yes
Different operationalisations of "Bird"	collapses to one number	assumes one referent	kept separate, requires explicit τ_{AB} to merge
Two surveys disagreeing in one context	collapses to mid-range strength with high confidence	depends on prior; typically collapses	flagged by f_{conflict}
Fly(Tweety) disputed by observers	collapses	requires explicit observer-noise model	stored as Both atom with provenance
Cognitive trigger to refine the ontology	none	none in the formalism	live high (n^+, n^-)-balance triggers refinement

Table 1: What survives, by representation strategy, in the Birds/Penguins worked example. The scalar STV column reflects a classical PLN deployment without context indexing. The global-ν column reflects what a νPLN-style system can do *in principle*, given a sufficiently expressive model class and a well-tuned prior; in many of these rows the information is recoverable but only via additional structure that the representation does not by itself supply. The πPLN column reflects the behaviour of the persistent evidence metagraph as specified in Sections 7–8.

Why this is what paraconsistency means here

The word "paraconsistency" is sometimes read as a licence to tolerate contradictions without doing anything about them. The worked example shows that the operationally meaningful reading is different. Paraconsistency in πPLN means three things, all of which are exhibited above:

1. Contradictory evidence does not cause classical explosion; the system does not infer $\perp$ or arbitrary ψ from co-present positive and negative support for ϕ. This is the negative requirement.
2. Contradictory evidence is preserved with provenance and context, so that downstream reasoning can localise it, schedule investigation, and sometimes refine the ontology that produced it. This is the positive requirement.
3. The persistent representation distinguishes high-and-balanced evidence (live conflict, $n^+ \approx n^-$ both large) from low-and-balanced evidence (ignorance, $n^+ \approx n^-$ both small) and from one-sided evidence (consensus, only one component large). Scalar projections collapse the first two; (n^+, n^-) does not.

The Birds/Penguins case is small enough that all three behaviours can be verified by hand. Scaling to a full metagraph of millions of statements with heterogeneous provenance, the same three structural properties are what allow a πPLN system to accumulate experience over decades without either losing nuance to averaging or losing coherence to contradiction.

16.2 Scope-shift contexts: cosmic, everyday, introspective

The Birds/Penguins example concerned specialisation within a shared ontology: penguins and ordinary birds are different reference classes, but both are classes of physical animals counted in the same sense. A more characteristic — and more practically pressing — application of πPLN is to cases where the contexts themselves carve the world differently, so that the same syntactic statement is supported by radically different (n^+, n^-) pairs and, in the limit, refers to subtly different things in different contexts. This is the situation an AGI system actually faces in ordinary use.

Scale-of-reference-class: a face is present

Consider the statement

$$\phi = \text{“a face is present in the immediate environment”}.$$

This is a perfectly ordinary observable predicate. Yet it has dramatically different empirical rates across reference classes that an embedded reasoner might plausibly invoke.

Let X_C denote the universe of discourse declared by context C — operationally, the kind of object that “a sample” refers to. For four contexts of practical interest:

- C_{cosmos}: X_C = cubic-metre regions of the observable universe, sampled by volume. Faces require evolved biological substrate on the surface of a habitable planet. Generous estimates of the mass-energy fraction of the universe in such substrate sit somewhere below 10^{-30}; the volume fraction is smaller still. Treating this as evidence that has been “observed” through cosmological modelling and inserting illustrative numbers:

$$(n^+_{C_{\text{cosmos}}}, n^-_{C_{\text{cosmos}}}) \sim (10^{10}, 10^{40}), \qquad s_{\text{emp}} \sim 10^{-30}.$$

- $C_{\text{urban-daytime}}$: X_C = five-second intervals of an urban human's waking life. Faces appear in person, on screens, in photographs and reflections. A typical week contains $\sim 5 \cdot 10^4$ such intervals, of which one might plausibly have face presence in $\sim 3 \cdot 10^4$:

$$(n^+_{C_{\text{urban}}}, n^-_{C_{\text{urban}}}) \sim (3 \cdot 10^4, 2 \cdot 10^4), \qquad s_{\text{emp}} \sim 0.6.$$

- $C_{\text{solo-hiking}}$: X_C = five-second intervals of solo wilderness hiking. Faces are encountered rarely:

$$(n^+_{C_{\text{hike}}}, n^-_{C_{\text{hike}}}) \sim (50, 10^4), \qquad s_{\text{emp}} \sim 5 \cdot 10^{-3}.$$

- $C_{\text{introspection}}$: X_C = moments of inward attention. The predicate “a face is present” is applicable only to imagined or remembered faces, which is a category the other three contexts do not invoke. Strict empirical counts in the original sense are not even defined here.

The same syntactic statement therefore lives in the persistent metagraph as

$$\begin{aligned}
K_t(\phi, C_{\text{cosmos}}) &= ((10^{10}, 10^{40}), \mathrm{Prov}_1, \ldots), \\
K_t(\phi, C_{\text{urban}}) &= ((3 \cdot 10^4, 2 \cdot 10^4), \mathrm{Prov}_2, \ldots), \\
K_t(\phi, C_{\text{hike}}) &= ((50, 10^4), \mathrm{Prov}_3, \ldots), \\
K_t(\phi, C_{\text{introspection}}) &= (\text{domain mismatch}),
\end{aligned}$$

with projected strengths spanning more than thirty orders of magnitude. None of these entries is wrong. They answer different questions.

Which probability does the query want?

Suppose a user asks an AGI assistant: *"How likely is it that I'm looking at a face right now?"* The query underspecifies the reference class. A πPLN chainer must do something a global-ν chainer cannot naturally do: *select the context whose reference class matches the operative meaning of the query.* Among the four contexts above:

- If the user is in their kitchen, the operative context is C_{urban} (or a refinement of it), giving $s \approx 0.6$.
- If the user is hiking alone on a ridge, the operative context is C_{hike}, giving $s \approx 0.005$.
- If the user is doing a cosmological order-of-magnitude estimate, the operative context is C_{cosmos}, giving $s \approx 10^{-30}$.
- If the user is asking about mental imagery, the operative context is $C_{\text{introspection}}$, which requires a different predicate altogether.

The chart selector reads task context, situational evidence (location, sensor input, recent dialogue), and the user's apparent register to pick the relevant C. This is everyday context disambiguation, and it is not typically modelled as Bayesian conditioning inside a master probability space. It is closer to a translation of an under-specified query into a context-tagged query, and that translation step is exactly what the πPLN chart-selection layer does.

A would-be global-ν system has two unattractive options. It can pick one canonical reference class — say, C_{cosmos} as the most "objective" — and treat every other rate as a conditional probability. This is technically possible but operationally backwards: most queries do not live in the cosmic context and would have to be expressed as deeply nested conditionals. Alternatively, the system can attempt to learn a hierarchical model that recovers each subclass rate as a posterior. This works in principle but requires the model class $\mathcal{L}$ to contain exactly the partition that the user's query implicitly invokes — which is a recursive instance of the same context-selection problem.

In πPLN the partition is up front, in the assumption packages. The chart selector does the disambiguation in one step rather than embedding it in the prior over $\mathcal{L}$.

Incommensurable ontologies: a mind is present

The face example involves contexts whose universes differ in size and sampling but whose predicates are mutually comprehensible (a face in the cosmos is the same kind of object as a face in a kitchen). A sharper case arises when the contexts disagree about what counts as an entity at all.

Let

$$\psi = \text{"a mind is present in the immediate vicinity"}.$$

Four contexts an AGI system regularly has to navigate:

- C_{phys}: a strict physicalist context whose universe is physical configurations and whose predicates range over biological substrate. "Mind" is operationalised as "brain executing certain functional processes." Empirical rate is bounded above by the volume fraction of nervous tissue in the universe, again astronomically small:

$$(n^+_{C_{\text{phys}}}, n^-_{C_{\text{phys}}}) \sim (10^{10}, 10^{40}).$$

- C_{social}: an everyday folk-psychological context whose universe is conversational partners and whose predicates include intention, belief, attention, mood. "Mind" is operationalised by theory-of-mind attribution. In any conversational exchange, minds are presumed on both sides:
$$(n^+_{C_{\text{social}}}, n^-_{C_{\text{social}}}) \sim (10^6, 10).$$
- C_{phen}: a phenomenological context whose universe is moments of conscious experience and whose primitives are qualia, intentionality, and attention. "Mind is present" is, in this context, not far from a tautology of being awake:
$$(n^+_{C_{\text{phen}}}, n^-_{C_{\text{phen}}}) \sim (\infty, 0)$$
(or, finitised, an arbitrarily large positive count against an empirical zero negative count).
- C_{anatta}: a deep-introspective context in the tradition of contemplative reports in which sustained inquiry into the felt sense of self reveals dissolution, absence, or radical impermanence. "Mind" here is operationalised as "a unified phenomenal subject," and the felt evidence runs the other way:
$$(n^+_{C_{\text{anatta}}}, n^-_{C_{\text{anatta}}}) \sim (10, 200).$$

These contexts disagree about ψ, but they do not just disagree *about a number*. They disagree about what ψ refers to. The predicate symbol "mind" picks out different referents in different $\mathcal{L}_C$. C_{phys} and C_{phen} are not straightforwardly related by Bayesian conditioning, because the conditioning event "the substrate is biological" lives in C_{phys} and is opaque to the predicates of C_{phen}. The translation maps

$$\tau_{C_{\text{phys}},C_{\text{phen}}}, \quad \tau_{C_{\text{phen}},C_{\text{anatta}}}, \quad \tau_{C_{\text{social}},C_{\text{phys}}}$$

are partial. Some have settled translations (the social-to-physical map is largely worked out by neuroscience, though contested at the margins). Others are open philosophical problems (the physical-to-phenomenal map is the hard problem of consciousness, and no settled τ exists). πPLN does not pretend to resolve these. It simply does not require their resolution in order to store, retrieve, and use evidence within each context.

The corresponding global-ν difficulty

What would a νPLN-style global probability space have to do here? In principle, the model class $\mathcal{L}$ would need to contain probabilistic programs whose outputs simulate all four contexts: a program for each $\mathcal{L}_C$, a meta-program for selecting between them, and a meta-prior over the meta-programs. All of this is mathematically expressible. But two practical points follow.

First, the meta-prior over context-selection programs is doing the same cognitive work as the πPLN chart selector. It is not avoided by embedding it in ν; it is hidden inside the prior. Making it explicit, as πPLN does, allows it to be inspected, debugged, and updated.

Second, the assumption that all four context-programs are commensurable elements of one $\mathcal{L}$ is a substantive claim — the claim that some unified probabilistic programming language can represent both the third-person physical predicates of C_{phys} and the first-person phenomenological predicates of C_{phen}. Whether such a unified $\mathcal{L}$ exists is a research question (it is, in effect, the hard problem of consciousness restated as a representation problem). Until it is answered, the global-ν formulation depends on an assumption that may not be available. πPLN does not depend on it: each context carries its own $\mathcal{L}_C$, and inter-context relations are partial maps.

Practical AGI behaviour under scope-shift contexts

An AGI assistant fielding realistic user queries routinely traverses scope shifts. Three patterns recur:

1. *Context inference from situational evidence.* "How rare is this?" from a hiker on a ridge is a C_{hike} query; the same words from a cosmologist at a conference are a C_{cosmos} query. The system uses dialogue, location, prior task context, and conversational register to pick the chart. Mistakes here are not numerical errors; they are category errors.

2. *Refusal to aggregate across incommensurable contexts.* "Is my sadness real?" admits a C_{phen} reading (where the answer is trivially yes), a C_{social} reading (where the answer is about whether others recognise it), and a C_{phys} reading (where the answer concerns neurochemical correlates). A helpful assistant does not average these; it identifies the operative reading and may surface the others if relevant. πPLN's context-indexed update law makes this behaviour structural rather than ad hoc.

3. *Translation as an explicit, fallible move.* When a user asks "in physical terms, what is happening when I feel sad?", they are requesting a partial $\tau_{C_{\text{phen}},C_{\text{phys}}}$ map. The system supplies what is known (neural correlates, transmitters, behavioural markers) while marking the residue (the felt quality itself) as untranslated. The persistent metagraph stores both: C_{phen} evidence about sadness, C_{phys} evidence about its correlates, and a partial-translation record between them. The user's eventual integration is a third context they construct over time.

In each pattern, the cognitive work the system does is not best modelled as posterior updating in one master probability space. It is context identification, context-indexed evidence retrieval, and explicit handling of partial translations. These are first-class operations in πPLN.

Revision under context mis-selection

Scope-shift examples are especially important for revision because the first error is often not a wrong number but a wrong context. Suppose the user asks "How likely is it that I am looking at a face right now?" while sitting alone in a dark room using a meditation app that periodically displays remembered faces as mental imagery. A shallow chart selector initially chooses C_{urban} and projects $s \approx 0.6$. A few seconds later, sensor and dialogue evidence arrive:

$$e_1 = \text{NoCameraFaceDetected}, \qquad e_2 = \text{UserReportsMentalImage}, \qquad e_3 = \text{EyesClosed}.$$

This should not merely decrement the urban face count. It should revise the query interpretation and chart selection.

Let H be the current selection state $C_* = C_{\text{urban}}$, and let $\mathcal{K}_C(H, E)$ include candidates:

1. K_0: keep C_{urban} and add negative physical-face evidence;

2. K_1: switch to $C_{\text{solo-indoor}}$;

3. K_2: switch to $C_{\text{introspection}}$ and translate the predicate from physical face to imagined face;

4. K_3: create a covering context distinguishing physical presence from imaginal presence.

Project these candidates and the evidence packet into a lattice $\mathcal{P}_C$ containing propositions about sensor state, dialogue state, attentional mode, and predicate sense. Then compute

$$\mathrm{fit}_C(K_i; E) = w_C(\mathrm{eval}_{\mathcal{E}}(E)) \Rightarrow_C w_C(\mathrm{eval}_{\mathcal{K}}(K_i) \wedge \mathrm{eval}_{\mathcal{E}}(E)),$$

$$\mathrm{pres}_C(K_i; H) = w_C(\mathrm{eval}_{\mathcal{K}}(H)) \Rightarrow_C w_C(\mathrm{eval}_{\mathcal{K}}(K_i) \wedge \mathrm{eval}_{\mathcal{K}}(H)).$$

The covering-context revision K_3 usually wins for an assistant: it preserves what was right about the urban reading (the user was asking an everyday question, not doing cosmology), fits the new introspective evidence, and avoids falsely updating the physical-face statistics with mental-imagery samples. The reply may then be: "No physical face appears to be in front of you, but you may be attending to a remembered or imagined face." This is revision of chart selection and predicate sense, not global Bayesian conditioning.

Revision of partial translations for mind predicates

The "mind is present" example also supports a more sophisticated revision pattern. Suppose new neuroscience evidence links a reported phenomenological state p to a physical marker m with high reliability. The update target is not simply

$$K(\psi, C_{\mathrm{phys}}) \otimes K(\psi, C_{\mathrm{phen}}).$$

The target is the partial translation

$$\tau_{\mathrm{phen}\to\mathrm{phys}}(p) = m$$

and its reliability/context boundary. Candidate revisions include:

1. strengthen the translation for this specific marker only;
2. introduce a subtype of phenomenological reports for which the marker is reliable;
3. reject the translation because the report protocol differs from the physical experiment;
4. create a bridge chart that carries both first-person report and third-person measurement as separate evidence channels.

Residual preservation is crucial here. A candidate that collapses the whole phenomenological ontology into the physical ontology may fit the new marker but preserves too little of C_{phen} and C_{anatta}. A bridge chart with a narrow translation often has better conservative revision score: it integrates the new scientific evidence while leaving untranslated residue explicit. Thus πPLN can learn cross-context bridges incrementally without pretending that the hard problem has been solved by one posterior update.

Summary

The scope-shift case sharpens the central claim of the paper. In the Birds/Penguins example, πPLN's advantage was that the persistent (n^+, n^-) atlas avoided sample-mix artefacts and ontology smear within a single basic ontology. In the cosmic/everyday/introspective case, πPLN's advantage is structurally deeper: the contexts do not share an ontology, and no global probability space is currently known that could represent them all without smuggling in a context-selection prior that does the same work as the πPLN chart selector. An AGI that must reason about ordinary user queries — how rare something is, whether a mind is present, what something means — does this reasoning across scope shifts every minute. πPLN is the part of PLN that takes the resulting structure seriously rather than factoring it out.

16.3 Roosters and sunrise

Temporal predictive implication alone does not imply causation. A local predictive chart may strongly support

$$\mathrm{Crow}(\mathrm{rooster}, t) \Rightarrow_{\mathrm{predictive}} \mathrm{Sunrise}(t + \Delta).$$

But a causal chart should require additional mechanism, intervention, and weakness constraints, in the spirit of explicit causal-model reasoning [12]. πPLN can store predictive evidence without promoting it to causal evidence unless a causal chart supports that move.

Suppose an intervention experiment suppresses rooster crowing while sunrise still occurs. The new evidence E_{int} is not primarily evidence against the predictive chart; the predictive chart may remain useful in farm contexts. It is evidence against a causal chart

$$H_{\mathrm{causal}} : \mathrm{Crow}(\mathrm{rooster}, t) \Rightarrow_{\mathrm{causal}} \mathrm{Sunrise}(t + \Delta).$$

Candidate revisions include:

1. downgrade the causal link to a predictive link;
2. add an omitted common cause, such as dawn light triggering crowing;
3. quarantine the causal inference trail that promoted prediction to causation;
4. revise the causal chart's admissibility rules to require intervention support.

A residuated revision rule selects among them by maximizing

$$(\mathrm{fit}(K; E_{\mathrm{int}}), \mathrm{pres}(K; H_{\mathrm{causal}}), w(K)).$$

The natural weakest adequate revision is usually not "rooster crowing is anti-causal." It is "the old correlation remains predictive in some contexts, but the causal promotion was invalid." This is corrective revision on the inference trail, not mere addition of negative evidence to a single scalar truth value.

16.4 Concept boundaries

For a vague predicate such as $\mathrm{Tall}(x)$ or $\mathrm{HighTemperature}(x)$, different contexts define different boundary conventions. A global probability space over one fixed predicate boundary is often the wrong object. πPLN stores context-indexed positive and negative evidence and only projects to a single scalar truth value for a particular decision.

Suppose C_A uses a strict basketball context where

$$\mathrm{Tall}_A(x) \equiv \mathrm{Height}(x) > 200\mathrm{cm},$$

while C_B uses an everyday context where

$$\mathrm{Tall}_B(x) \equiv \mathrm{Height}(x) > 180\mathrm{cm}.$$

Evidence that Bob is 185cm is positive evidence for $\mathrm{Tall}_B(\mathrm{Bob})$ and negative evidence for $\mathrm{Tall}_A(\mathrm{Bob})$. These are not contradictory measurements; they are different concept boundaries. Revision is guarded by translation:

$$\tau_{B \to A}(\mathrm{Tall}_B(x))$$

is not the identity map unless the system explicitly constructs a covering context that relates the two thresholds. A decision context C_D can revise a joint assessment by choosing translations $\tau_{A\to D}$ and $\tau_{B\to D}$ that maximize residual fit to the decision evidence while preserving each local boundary. Without these translations, no automatic evidence aggregation occurs.

This example illustrates why context-indexed revision is not a technical luxury. It prevents the metagraph from smearing strict and loose meanings into one incoherent average.

16.5 Synthetic non-gluable charts

A more diagnostic synthetic benchmark uses locally coherent pairwise probability charts that agree on all overlaps but still do not extend to a global joint distribution. Let X, Y, Z be $\{-1, +1\}$-valued variables. The charts C_{XY}, C_{YZ}, and C_{XZ} each use the same unbiased pairwise anti-correlation marginal:

$$P(++) = P(--) = \frac{1}{16}, \qquad P(+-) = P(-+) = \frac{7}{16}.$$

Every single-variable marginal is uniform, so the charts are compatible on overlaps. But the implied pairwise correlation is $r = -3/4$ for all three pairs, and the corresponding correlation matrix has eigenvalue $1 + 2r = -1/2$. No global probability distribution exists.

Revision can be posed as follows. Let H be the current atlas containing the three pairwise charts, and let E be a task requiring a single joint answer about (X, Y, Z). Candidate revisions include:

1. demote one pairwise chart;
2. weaken all three anti-correlations until the correlation matrix becomes positive semidefinite;
3. create a covering context with an explicit contextuality/non-gluability marker;
4. keep the charts separate and answer that the task is ill-posed without a choice of context.

The hard revision rule may impose a fit threshold requiring a single joint model, in which case some anti-correlation must be weakened. The soft revision rule may instead prefer the weakest adequate response: preserve all three local charts and report non-gluability as the answer. In the lattice-pullback notation,

$$\mathrm{pres}(K; H) = w(\mathrm{eval}_{\mathcal{K}}(H)) \Rightarrow w(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{K}}(H))$$

measures how much local structure survives the revision. This is a clean algebraic account of "gluing as achievement, not axiom."

The older Boolean triangle

$$A = B, \qquad B = C, \qquad A \neq C$$

remains useful as a toy logical cartoon. The anti-correlation example is the probabilistic version that matters for evaluating global-distribution claims.

16.6 Source correction in biomedical inference

Suppose a biomedical PLN chainer combines several abstracts to infer

$$\phi = \text{DrugA reduces MarkerB in ContextC}.$$

Three documents support ϕ, but two of them are later discovered to cite the same flawed experimental dataset. A scalar Bayesian update might add a new negative likelihood factor or lower the posterior. πPLN first performs corrective revision on provenance:

$$\mathcal{E}_{t+1}(\phi, C) = \mathrm{Corr}(\mathcal{E}_t(\phi, C), P_{\text{flawed}}, \eta).$$

The flawed packets may be discounted or quarantined rather than treated as negative evidence for $\neg\phi$. The remaining independent support is then reprojected to an STV or beta posterior if a local decision chart requires one.

If a new high-quality randomized trial supplies genuine counterevidence, that is a different revision operation:

$$K_{t+1}(\phi, C) = K_t(\phi, C) \otimes (0, m^-),$$

possibly with high reliability. The system can therefore distinguish "support was duplicated or invalid" from "there is new evidence against the claim." This distinction is crucial in scientific reasoning and cannot be cleanly represented by merely revising a collapsed scalar probability.

16.7 Program-patch revision

Let h be a learned rule or program used by PLN inference control, and let $B_h \subseteq X \times Y$ be its behavior relation. A new counterexample $e = (x_e, y_e)$ induces evidence relation E_e. A local patch search proposes candidate programs h' with behavior $B_{h'}$. The revision rule computes

$$\mathrm{fit}(h'; e) = w(\mathrm{eval}_{\mathcal{E}}(E_e)) \Rightarrow w(\mathrm{eval}_{\mathcal{K}}(B_{h'}) \wedge \mathrm{eval}_{\mathcal{E}}(E_e)),$$

$$\mathrm{pres}(h'; h) = w(B_h) \Rightarrow w(B_{h'} \cap B_h).$$

A brittle special-case patch may have high fit but low weakness and low reuse. A broader patch may repair the counterexample while preserving more old behavior and making fewer new distinctions. The selected program maximizes

$$(\mathrm{fit}(h'; e), \mathrm{pres}(h'; h), w(B_{h'}), -\mathrm{Cost}(h')).$$

This example shows revision as generalized program repair: it is not Bayesian conditioning over a world distribution, but weakest adequate transformation of a local behavior relation.

17 Evaluation plan

A practical evaluation should compare ordinary PLN, global νPLN, and πPLN on tasks involving:

1. inconsistent local contexts;
2. default reasoning with exceptions;
3. vague concept boundaries;
4. causal inference under confounding;
5. biomedical multi-document inference;
6. background inference over large metagraphs;
7. factor-graph scheduling with local Bayesian islands;

8. revision under duplicated provenance, source correction, and stale evidence;
9. cross-context ontology revision where concept boundaries must be translated rather than averaged;
10. non-gluable-atlas revision where the correct answer may be to preserve local charts and report incompatibility.

Useful metrics include:

1. task success;
2. calibration;
3. conflict preservation;
4. false confidence inflation;
5. proof/search cost;
6. transfer across task variants;
7. quality of provenance;
8. revision preservation versus adaptation;
9. provenance-correction accuracy;
10. ontology-smear avoidance.

The expected advantage of πPLN is not always sharper scalar probability. It is better contextualization, less brittle inference, lower false confidence, and more useful handling of heterogeneous evidence.

18 Objections and replies

18.1 Objection: without a global distribution, the theory is less rigorous

Reply: local rigor is preserved. The system constructs explicit local probability spaces when needed. The difference is that global coherence is a theorem to be earned in special cases, not an axiom.

18.2 Objection: paraconsistency just tolerates inconsistency

Reply: πPLN does not merely tolerate inconsistency. It measures and localizes it. Conflict is stored as positive and negative evidence with provenance and context tags.

18.3 Objection: Bayesian reasoning requires a sample space

Reply: Bayesian reasoning requires a sample space for a given calculation. πPLN supplies such a sample space locally. It denies only that all calculations must share one master sample space.

18.4 Objection: this is just historical PLN context

Reply: historical PLN context is a precursor, but πPLN adds:

1. explicit local νPLN chart construction;
2. p-bit/count quantale semantics;
3. non-gluability diagnostics;
4. weakness-guided chart selection;
5. quantale-valued chainer labels;
6. local-Bayesian-island factor graphs.

18.5 Objection: proper revision requires a global distribution

If one defines proper revision as globally coherent Bayesian conditioning over every belief in the system, then this objection is correct by definition. But that is not the right definition for an AGI metagraph. Most practical revision concerns empirical evidence packets, provenance overlap, stale observations, defective sources, context translations, and ontology changes. These can be handled by the context-indexed revision rules of Section 11 without a global probability space.

The honest distinction is this: Bayesian posterior formation requires a prior; persistent empirical revision does not. πPLN stores raw counts and p-bits. When a local chart wants a posterior, it injects a local prior. The prior is part of the chart, not the global mind.

19 Conclusion

The νPLN formalism of [1] gives PLN a useful local probabilistic coordinate system. It shows how to derive familiar PLN truth-value calculations, especially beta posterior direct-introduction calculations, from a probability space over possible predicate-generating programs. πPLN keeps this mathematics but relocates it. A νPLN probability space is a local chart, not the ontology of the entire AGI system.

The enduring state of a πPLN system is a context-indexed, provenance-rich, paraconsistent evidence metagraph. Its primitive truth/evidence object is an empirical positive/negative count or p-bit object. PLN STVs, beta truth-value distributions, scalar probabilities, and decision-time plausibility scores are projections of this object under local assumptions. Priors are therefore chart-level parameters, not global semantic foundations.

The most important addition in this version of the paper is revision. πPLN can answer the worry that revision requires a global distribution by separating three levels:

1. persistent empirical revision of p-bit/count evidence;
2. local Bayesian posterior revision inside a temporary chart;
3. higher-order revision of valuations, contexts, translations, and ontologies.

The general residuated rule is simple: fit the new evidence, preserve as much old high-value structure as possible, and choose the weakest adequate revised state. In probability-valued contexts this recovers conditional-mass and Bayesian-looking formulas. In count-valued contexts it recovers beta-binomial updating after projection. In paraconsistent and product-quantale contexts it handles conflict, provenance, cost, weakness, attention, and context translation without pretending that all evidence belongs to one master sample space.

For implementation, the punchline is direct. A chainer should not merely attach one truth value to an Atom. It should derive and revise context-provenanced p-bit evidence packets. A factor graph should not be understood as one global joint probability model. It should be a quantale control graph coordinating local Bayesian islands, rule factors, revision factors, weakness factors, provenance factors, and task-utility factors.

Thus:

$$\nu\text{PLN} = \text{local Bayesian coherence for PLN},$$

$$\pi\text{PLN} = \text{paraconsistent contextual intelligence using local Bayesian charts.}$$

The disagreement with the global-ν view is not over whether Bayesian mathematics is valid. It is over scope. Bayesian coherence is a powerful local achievement. A real AGI needs the atlas, transition maps, revision rules, provenance calculus, conflict preservation, weakness-guided chart selection, and inference-control machinery surrounding that achievement.

A Notation and preliminaries

A.1 PLN STVs

An STV may be written (s, n) or (s, c), where

$$s \in [0,1], \quad n \geq 0, \quad c = \frac{n}{n+k}.$$

In πPLN, the persistent evidence object is the empirical pair (n^+, n^-). The empirical count is

$$n = n^+ + n^-.$$

A prior-aware STV projection with prior strength p_0 and virtual count k is

$$s = \frac{n^+ + kp_0}{n+k}, \qquad c = \frac{n}{n+k}.$$

Conversely, if (s, c) is known to have been produced from empirical count n, prior strength p_0, and virtual count k, then

$$n^+ = s(n+k) - kp_0, \qquad n^- = n - n^+,$$

provided the right-hand side lies in $[0, n]$. The prior-free empirical projection is the special case $k = 0$:

$$s_{\text{emp}} = \frac{n^+}{n^+ + n^-}.$$

A.2 Beta distribution

The beta distribution $\text{Beta}(\alpha, \beta)$ has density proportional to

$$\theta^{\alpha-1}(1-\theta)^{\beta-1}.$$

With Bernoulli observations containing n^+ successes and n^- failures, a beta prior $\text{Beta}(\alpha_0, \beta_0)$ updates to

$$\mathrm{Beta}(\alpha_0 + n^+, \beta_0 + n^-).$$

A.3 Quantale-valued relations

Given a commutative quantale Q, a Q-valued relation $R : X \times Y \to Q$ composes with $S : Y \times Z \to Q$ by

$$(S \circ R)(x, z) = \bigvee_{y \in Y} R(x, y) \otimes S(y, z).$$

This is the basic algebra behind quantale factor-graph message passing and approximate context translation.

B Proofs

B.1 Proof of the evidence-count quantale theorem

We must show that

$$Q_{\pm}^{\mathrm{cnt}} = ([0, \infty]^2, \leq, \vee, +, (0, 0))$$

is a commutative quantale.

The order is componentwise, so $[0, \infty]^2$ is a complete lattice with componentwise suprema. Addition is associative and commutative componentwise, with unit $(0, 0)$. Finally, for any family $\{b_i\}$ and any a,

$$a + \bigvee_i b_i = \bigvee_i (a + b_i)$$

componentwise, because addition by a fixed extended nonnegative number preserves suprema. Hence tensor distributes over arbitrary joins.

B.2 Proof of the bounded accumulation quantale theorem

For one coordinate, define

$$a \star b = a + b - ab = 1 - (1 - a)(1 - b).$$

The operation $\star$ is associative and commutative because multiplication of $(1 - a)$ factors is associative and commutative. Its unit is 0, since $a \star 0 = a$. It preserves arbitrary suprema in each argument because, for fixed a, the map

$$b \mapsto a \star b = a + (1 - a)b$$

is monotone and sup-preserving on $[0, 1]$. Therefore $([0, 1], \leq, \sup, \star, 0)$ is a commutative quantale, and the componentwise product of two copies is also a commutative quantale.

For the homomorphism claim, compute in one coordinate:

$$\rho_\lambda(n_1) \star \rho_\lambda(n_2) = (1 - e^{-n_1/\lambda}) \star (1 - e^{-n_2/\lambda})$$

$$= 1 - e^{-(n_1+n_2)/\lambda} = \rho_\lambda(n_1 + n_2).$$

The two-coordinate tensor statement follows componentwise. The join statement also follows componentwise because ρ_λ is monotone and therefore preserves componentwise suprema:

$$\rho_\lambda(\bigvee_i n_i) = \bigvee_i \rho_\lambda(n_i).$$

Hence bounded propagation is a strict quantale image of evidence-count accumulation.

B.3 Proof of the STV projection theorem

Given empirical evidence counts (n^+, n^-), let $n = n^+ + n^-$. A context prior with expected strength p_0 and virtual count k is equivalent to virtual positive and negative counts

$$\alpha_0 = kp_0, \qquad \beta_0 = k(1 - p_0).$$

The posterior expected strength after adding empirical counts is therefore

$$s = \frac{\alpha_0 + n^+}{\alpha_0 + \beta_0 + n} = \frac{n^+ + kp_0}{n + k}.$$

The PLN confidence projection

$$c = \frac{n}{n + k}$$

records the proportion of total virtual-plus-empirical count that comes from empirical evidence. Thus (n^+, n^-) plus the context prior (p_0, k) determines the STV (s, c). For $k = 0$, this reduces to the empirical maximum-likelihood strength n^+/n.

B.4 Proof of local beta recovery

Inside a Bernoulli chart, likelihood for θ from n^+ positive and n^- negative observations is

$$L(\theta) = \theta^{n^+}(1 - \theta)^{n^-}.$$

The prior density is proportional to

$$\theta^{\alpha_0 - 1}(1 - \theta)^{\beta_0 - 1}.$$

Multiplying prior and likelihood yields a posterior density proportional to

$$\theta^{\alpha_0 + n^+ - 1}(1 - \theta)^{\beta_0 + n^- - 1},$$

which is the density of

$$\mathrm{Beta}(\alpha_0 + n^+, \beta_0 + n^-).$$

This is exactly the beta posterior used in the predicate direct introduction derivation of [1], with $n^+ = n$ and $n^- = N - n$.

B.5 Proof of non-explosion

πPLN base semantics stores evidence values (n^+, n^-) and does not include the classical explosion rule

$$\phi, \neg\phi \vdash \psi.$$

A state with both $n^+_{\phi,C} > 0$ and $n^-_{\phi,C} > 0$ records support and opposition for ϕ in context C. If support and opposition live in different contexts, C_1 and C_2, they remain separated until a translation or covering context is explicitly constructed. No rule schema licenses arbitrary ψ from either fact alone. Therefore contradiction is non-explosive by construction. Classical behavior is recovered only under a projection or sublogic that explicitly imposes classical consistency.

B.6 Proof of the probabilistic non-gluability theorem

Each pairwise chart is a valid probability distribution: its four probabilities are nonnegative and sum to one. The one-variable marginals are uniform because in each pairwise chart, marginalizing out one variable gives $P(\cdot = +1) = P(\cdot = -1) = 1/2$ for the remaining variable. Thus the pairwise charts agree on all single-variable overlaps.

The expected product in each pair is

$$E[XY] = P(X = Y) - P(X \neq Y) = \left(\frac{1}{16} + \frac{1}{16}\right) - \left(\frac{7}{16} + \frac{7}{16}\right) = -\frac{3}{4},$$

and analogously for $E[YZ]$ and $E[XZ]$. If a joint distribution on (X, Y, Z) existed, the matrix of second moments

$$R = \begin{pmatrix} 1 & r & r \\ r & 1 & r \\ r & r & 1 \end{pmatrix}, \qquad r = -\frac{3}{4},$$

would be positive semidefinite, because for any vector a, $a^T Ra = E[(a_1X + a_2Y + a_3Z)^2] \geq 0$. But the eigenvalues of this equicorrelation matrix are $1 - r, 1 - r, 1 + 2r$, and $1 + 2r = -1/2 < 0$. Therefore R is not positive semidefinite, contradiction. Hence no global joint distribution has the three specified pairwise marginals, despite their local coherence and overlap compatibility.

B.7 Proof of the product quantale theorem

Let $Q_i = (Q_i, \leq_i, \bigvee_i, \otimes_i, e_i)$ be commutative quantales for $i = 1, \ldots, m$. Define the product order, product joins, product tensor, and product unit componentwise. Completeness follows from componentwise completeness. The monoid laws follow componentwise. Distributivity follows because, for each component,

$$a_i \otimes_i \bigvee_j b_{ij} = \bigvee_j (a_i \otimes_i b_{ij}).$$

Thus the product is a commutative quantale.

B.8 Proof sketch of the conservative residuated revision rule

Let Q_C be a commutative residuated quantale. The residual is characterized by

$$a \otimes b \leq c \quad \Longleftrightarrow \quad b \leq (a \Rightarrow c).$$

Candidates, evidence packets, and previous hypotheses may have different native types, so the revision rule first projects them into a shared complete lattice $\mathcal{P}_C$ by

$$\mathrm{eval}_{\mathcal{K}} : \mathcal{K}_C(H, E) \to \mathcal{P}_C, \qquad \mathrm{eval}_{\mathcal{E}} : \mathcal{E}_C \to \mathcal{P}_C.$$

The meet $\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E)$ is therefore a well-typed coverage object: the part of the candidate's local logical extension that matches the evidence. Thus

$$w_C(\mathrm{eval}_{\mathcal{E}}(E)) \Rightarrow_C w_C(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E))$$

is the greatest residual factor that can be combined with the evidence value while remaining below the weakness value of the part of the revised state covering that evidence. This justifies reading it as a fit gap or fit score. The same argument gives the preservation score

$$w_C(\mathrm{eval}_{\mathcal{K}}(H)) \Rightarrow_C w_C(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{K}}(H)).$$

The hard rule

$$\mathrm{Rev}_{C,\tau}(H, E) = \underset{K:\mathrm{fit}_C(K;E)\geq\tau}{\overset{\mathrm{lex}}{\arg\max}} \left(\mathrm{pres}_C(K; H), w_C(\mathrm{eval}_{\mathcal{K}}(K))\right)$$

therefore enforces fit first by the constraint, then chooses the revision that preserves old high-value structure, and finally chooses the weakest among preservation-equivalent candidates. The soft rule simply moves fit into the lexicographic objective. No global probability distribution is used in the argument; only the local proposition lattice, quantale, candidate generator, and weakness valuation are needed.

B.9 Proof sketch of the probability-valued revision special case

Let $Q = ([0, 1], \leq, \times, 1, \sup, \Rightarrow)$ with residual

$$a \Rightarrow b = \begin{cases} 1, & a \leq b, \\ b/a, & 0 < b < a, \\ 0, & b = 0 < a. \end{cases}$$

If $\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E) \leq \mathrm{eval}_{\mathcal{E}}(E)$, then $w(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E)) \leq w(\mathrm{eval}_{\mathcal{E}}(E))$. Hence, for $w(\mathrm{eval}_{\mathcal{E}}(E)) > 0$,

$$\mathrm{fit}(K; E) = w(\mathrm{eval}_{\mathcal{E}}(E)) \Rightarrow w(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E)) = \frac{w(\mathrm{eval}_{\mathcal{K}}(K) \wedge \mathrm{eval}_{\mathcal{E}}(E))}{w(\mathrm{eval}_{\mathcal{E}}(E))}.$$

This is precisely the conditional mass of the portion of evidence covered by the candidate. The same calculation gives preservation as a conditional mass of old hypothesis preserved by the revised state. Therefore the residuated revision schema reduces to conservative conditional-mass revision in the local probability quantale.

B.10 Proof sketch of count revision and beta projection

Same-context empirical count revision uses the tensor of $Q_{\pm}^{\mathrm{cnt}}$:

$$(n^+, n^-) \otimes (m^+, m^-) = (n^+ + m^+, n^- + m^-).$$

Projecting after revision gives

$$\Pi_{\mathrm{Beta}}((n^+, n^-) \otimes (m^+, m^-)) = \mathrm{Beta}(\alpha_0 + n^+ + m^+, \beta_0 + n^- + m^-).$$

If one first projects (n^+, n^-) to

$$\mathrm{Beta}(\alpha_0 + n^+, \beta_0 + n^-)$$

and then performs a beta-binomial update with m^+ successes and m^- failures, conjugacy yields the same posterior. Thus empirical count revision commutes with local beta projection for independent Bernoulli evidence.

B.11 Proof sketch of overlap-aware count discounting

Let two evidence packets have effective counts n_1, n_2 and overlap coefficient $c \in [0, 1]$. The classical PLN-inspired discounted count

$$n = n_1 + n_2 - c \min(n_1, n_2)$$

has the desired boundary behavior. If $c = 0$, the packets are treated as disjoint and $n = n_1 + n_2$. If $c = 1$ and one packet is fully redundant with the other, the resulting count is $\max(n_1, n_2)$, avoiding false inflation. Since

$$\max(n_1, n_2) \le n \le n_1 + n_2,$$

the formula interpolates monotonically between full redundancy and full independence. Applying the same argument componentwise to positive and negative counts gives the simple overlap-aware πPLN update. More refined versions replace c with a provenance- or dependency-estimated matrix, but the same boundary argument applies.

B.12 Proof sketch of guarded cross-context revision

Let $\tau_{D\to C}$ be a partial translation of statements and let $r_{D\to C} \in Q_{\pi\mathrm{PLN}}$ be its reliability. Transported evidence is admissible only when $\tau_{D\to C}$ is defined and its reliability passes the context's threshold. If it is not defined, no update to $K(\phi, C)$ is made; therefore automatic ontology smear is impossible by construction. If it is defined, the transported packet is explicitly marked with translation provenance and combined using $\otimes_{\mathrm{Prov}}$, so later correction or quarantine can act on the translation itself. Hence cross-context revision is guarded rather than background aggregation.

B.13 Proof of the lax probability valuation claim

Let $\mathcal{E}$ be an event algebra with tensor $\cap$. A strict homomorphism from $(\mathcal{E}, \cap, \Omega)$ to $([0, 1], \times, 1)$ would require

$$\nu(A \cap B) = \nu(A)\nu(B)$$

for all events A, B. This is exactly mutual independence for all pairs of events, which is not true in general probability spaces. Therefore a probability valuation is not generally such a homomorphism. It becomes one only on independent subfamilies or under explicit factorization assumptions.

B.14 Proof sketch of rule lifting

Let r be a PLN rule with local truth-value function f_r^C. In a fixed chart C, the premises ϕ_i have local truth values $\mathrm{TV}_C(\phi_i)$. Applying the rule gives $\mathrm{TV}_C(\psi)$. A projection map Π_C sends $\mathrm{TV}_C(\psi)$ to an evidence increment (Δ_C^+, Δ_C^-). Therefore

$$\widehat{f_r} = \Pi_C \circ f_r^C$$

is a well-defined lifted rule from persistent evidence plus context to a persistent evidence update. Provenance is added as metadata.

B.15 Proof sketch of chainer local soundness

A πPLN chainer step records its rule, premises, target, context, and chart. By construction, any truth-value calculation performed by the step is performed inside the recorded chart Γ_C. Therefore the conclusion is sound relative to that chart's assumptions and local semantics. The global system does not assert chart-independent soundness; it asserts chart-provenanced local soundness.

B.16 Proof sketch of quantale belief propagation on trees

On an acyclic factor graph, removing an edge separates the graph into subtrees in the algebraic sense of dynamic programming. The message from a subtree to its boundary variable is the join over all internal assignments of the tensor product of internal factors.

Crucially, these messages are elements of the evidence/control quantale, such as $Q_{\pm}^{\mathrm{cnt}}$ or $Q_{\pi\mathrm{PLN}}$, not elements of a local event algebra Q_C^{ev} and not global probabilities. Local probability valuations are generally lax because variables may be dependent and because $\nu_C(A \cap B) = \nu_C(A)\nu_C(B)$ holds only under local independence assumptions. That laxness is quarantined inside local Bayesian islands.

The global message-passing graph routes evidence. In $Q_{\pm}^{\mathrm{cnt}}$, tensor is addition of evidence counts and join is componentwise supremum, so the strict distributivity law holds:

$$a \otimes (b \vee c) = (a \otimes b) \vee (a \otimes c).$$

The same holds componentwise for the product control quantale, by the product quantale theorem. By induction from leaves to root, the recursive message equations compute exactly these subtree summaries. The final belief at each variable is therefore the exact quantale evidence marginal: the maximum or least-upper-bound supportable evidence summary across the relevant proof or assignment alternatives. It is not claimed to be a global joint probability marginal.

B.17 Proof sketch of the local Bayesian island theorem

Within a local chart C, variables may be highly dependent. The strict distributivity needed for global quantale routing need not hold for the local probability valuation ν_C viewed as a map from event intersection to multiplication. Therefore ordinary probabilistic inference, such as sum-product, beta-binomial updating, or another chart-appropriate method, is used internally to compute local marginals or sufficient statistics under the explicit joint/factorized model of C.

For Bernoulli or binomial local truth values, these sufficient statistics are empirical counts (n_C^+, n_C^-). The export map sends them into $Q_{\pm}^{\mathrm{cnt}}$, along with context, assumptions, and provenance. Once exported, aggregation across charts uses the paraconsistent evidence quantale, not a global joint distribution. By quarantining dependent probability calculation inside local Bayesian islands, mathematically ordinary local Bayesian inference and globally strict paraconsistent evidence propagation coexist without algebraic contradiction.

B.18 Proof sketch of MDL/MAP-under-universal-prior as a weakness special case

Choose hypotheses to be programs and choose a weakness score monotone in negative program length:

$$w(H) = 2^{-K(H)}.$$

Maximizing fit times weakness, or minimizing cost subject to fit, becomes the usual MDL or MAP-under-universal-prior preference for shorter programs. Solomonoff induction proper mixes over all programs weighted by the universal prior; the arg max rule here recovers its MAP/MDL face, not the full mixture. Therefore description-length induction is a special case of weakness-guided selection under a particular quantale and coding scheme.

C Implementation sketch

C.1 Data carried by an inferred Atom

A practical πPLN Atom should carry at least:

$$\begin{aligned}&(\text{Statement}, (n^+, n^-), \text{STVProjection},\\ &\ \text{ContextID}, \text{ChartID}, \text{Assumptions},\\ &\ \text{Provenance}, \text{Weakness}, \text{TaskTrace}, \text{Attention}).\end{aligned}$$

The scalar STV is cached for compatibility. The evidence-count pair and provenance are primary.

C.2 Chainer pseudocode

```
piPLN-backward-chain(target phi, task T, budget B):

    agenda := candidate_contexts(phi, T, K)
    return_packets := empty_provenance_join()

    while budget remains and agenda not empty:

        item := pop_best(agenda)

        C := item.context
        Gamma := retrieve_or_build_chart(C, K, T)

        expansions := expand_PLN_rules(phi, C, Gamma)

        for expansion in expansions:

            local_tv := evaluate_inside_chart(expansion, Gamma)
```

```
        delta := project_to_evidence(local_tv, C)

        delta := provenance_correct(delta, expansion.provenance)

        delta := novelty_discount(delta, K(phi,C), expansion.provenance)

        K.update_guarded(phi, C, delta)

        return_packets := provenance_join(return_packets,
                                          (C, delta,
                                           expansion.provenance))

        new_items := propose_subgoals(expansion, C, Gamma)

        for ni in new_items:
            ni.priority := scalarize_quantale_label(ni.label, T)
            push(agenda, ni)

    return project_for_decision(K, phi, target_context,
                                return_packets)
```

Here `provenance_join` denotes union-with-deduplication on provenance-tagged packets, preserving context IDs and source trails rather than adding counts blindly.

The variable `return_packets` is not the authoritative truth value of ϕ. It is a provenance-separated summary of what this particular chainer run contributed. The authoritative state is still the context-indexed metagraph entry $K(\phi, C)$, updated through guarded provenance-aware revision. This avoids double-counting when multiple expansions of ϕ share subgoals, documents, or proof trails.

C.3 Factor-graph pseudocode

```
piPLN-factor-message(factor f, variable v):

    for each value xv of v:

        msg[xv] := bottom_Q

        for each assignment x_rest of neighbors(f) except v:

            term := factor_value_Q(f, xv, x_rest)

            for each u in neighbors(f) except v:
                term := term tensor message[u -> f][x_u]

            msg[xv] := msg[xv] join term

    return msg
```

When a factor is a local Bayesian island, its internal message calculation may be ordinary probabilistic sum-product. Its output is then projected to a quantale value before being sent to the surrounding πPLN graph.

D Suggested paper-to-implementation mapping

Paper object	Implementation object
Statement ϕ	Atom, MeTTa expression, or typed metagraph pattern
Context C	Context node or MeTTa `Space` (acting as a native topological boundary enforcing cross-context guarded revision and ontology separation), with language fragment, assumptions, evidence selector, and task tag
Chart Γ_C	Temporary probabilistic model or factor-graph subproblem
(n^+, n^-)	Persistent truth/evidence payload
STV projection	Cached compatibility truth value for ordinary PLN rules
Weakness quantale	Rule/chart scheduling score algebra
Product quantale label	Full metadata label on proof step or factor message
Provenance	Source graph, proof path, observation IDs, chart IDs
Revision packet	Evidence update with fit, preservation, context, source, and correction metadata
Revision factor	Factor-graph node implementing residual fit/preservation and guarded update
Corrective operation	Quarantine, discount, retraction, relabeling, or counterevidence generation over provenance packets
Conflict	High positive and high negative evidence with nontrivial provenance separation
Local Bayesian island	Ordinary probabilistic factor graph scoped to one context

E Bibliographic note

The present draft is written as a mathematical and conceptual riposte to the global-probability-space interpretation of νPLN. It relies on six main background documents: the νPLN draft [1], the PLN book [3], the paraconsistent foundations paper [2], the weakness theory manuscript [4], Hyperseed [5], and the associated local revision rules draft [6]. The core novelty here is the relocation of νPLN into local charts and the explicit quantale account of persistent paraconsistent evidence and inference control.

AI Acknowledgement

In the course of drafting, refining, and stress-testing the arguments and formalism in this paper, the author made substantial use of three contemporary large language model assistants: ChatGPT-5.5-Pro, Gemini Pro, and Claude Opus 4.7. These tools were used variously for sharpening definitions, surfacing counterexamples, suggesting clearer prose, checking the internal consistency of the quantale and revision algebra, and stress-testing the argument against likely objections from a global-probability-space perspective. All mathematical claims, conceptual choices, and final formulations

remain the responsibility of the author.

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